PHYS3661: Theoretical Physics 3

Quantum Theory 3

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Chapter 1

Scattering

1.1 Scattering experiments and cross sections

Core reading: Chapter 13.1 of Bransden & Joachain

1.1.1 History and Motivation

Atoms are built of electrons and nuclei. Electrons are elementary. Nuclei, however, consist of nucleons — *i.e.* protons and neutrons — and these are known to be complicated bound states made predominately of three quarks, which are also believed to be elementary. This picture of the world has been built up over the last 100 years. But how was it deduced?

Most of this knowledge comes from scattering experiments, such as the Rutherford experiment (Geiger and Marsden 1909, Rutherford 1911) or modern day facilities such as the Large Hadron Collider based in CERN. The corresponding scattering events take place on a microscopic level. Thus an understanding of the quantum mechanical foundations of scattering is the basis for a fundamental understanding of the building blocks of the universe.

1.1.2 Basics of scattering

In quantum mechanical scattering, we imagine a particle localised at $z = -\infty$ travelling towards the origin, where a target has been prepared. Interactions between the incoming and target particles occur, and we observe what flies out from the collision in all directions.

There are two types of scattering process that we need to understand: *elastic* and *inelastic*.

Elastic scattering:

A scattering process is called *elastic* when the incoming particles stay intact. In general, they may change direction and energy, but do not fragment or produce new particles.

For example:

Rutherford's experiment saw the scattering of α particles ($\alpha = {}^4_2\mathrm{He}^{2+} = ppnn$) from Gold nuclei,

$$\alpha + \mathrm{Au} \to \alpha + \mathrm{Au}.$$

The α particles emerged scattered in all directions, but they did not change into different particles. Nor did they appear accompanied by showers of new particles.

Inelastic scattering:

We define *inelastic* processes in contrast to elastic ones: a process is inelastic when it is not elastic. There are many types of scattering process where this occurs.

For example:

(a) In nuclear fission, an incoming neutron hits a large nucleus which then splits into smaller nuclei (along with further neutron emission)

$$n + {}^{235}\mathrm{U} \to {}^{92}\mathrm{Kr} + {}^{141}\mathrm{Ba} + 3n.$$

In this case, the Uranium nucleus has split into large fragments violating the definition of elastic scattering.

(b) At lower energies, a collision with an incoming particle can excite a composite particle into a higher energy state, *e.g.* an electron bumps into an atom, lifting one of its electrons out of the ground state,

$$e^- + \mathrm{A} \to e^- + \mathrm{A}^*.$$

The excited particle would then be expected to transition back to its ground state radiatively by the emission of a photon,

$$\mathrm{A}^* \to \mathrm{A} + \gamma.$$

(c) It is also possible for new particles to be created in a quantum scattering process. For example, at high energies electron positron colliders can produce heavier leptons such as muons (and anti-muons).

$$e^- + e^+ \to \mu^- + \mu^+.$$

1.1.3 Cross Section

In this course, we will focus on *non-relativistic elastic* scattering. Our main interest is in computing how likely it is for the incoming particle to be scattered in a given direction. This is characterised by a quantity called the *cross section*.

For an incoming flux of particles F and an observed rate of particles $R(\theta, \phi)$, the *differential cross section* is defined to be

$$\frac{d\sigma}{d\Omega} = \frac{1}{F}\frac{dR}{d\Omega}. \tag{1.1}$$

We can also compute the *total cross section* by integrating this over the sphere,

$$\sigma = \int d\Omega \, \frac{d\sigma}{d\Omega} = \int_o^{2\pi} d\phi \int_0^{\pi} \sin\theta d\theta \, \frac{d\sigma}{d\Omega}. \tag{1.2}$$

Looking at our definition, we see that the differential cross section is a normalised rate of particle scattering in a given direction. This gives us an idea of the likelihood of the process in question.

We can also compute the total number of scattering events,

$$N = \left(\int F dt \right) \sigma,$$

where the integral of the flux F gives us the total number of particles per unit area delivered to the target in our experiment. Dimensional analysis shows that σ has the units of area,

$$N = \left(\int F dt \right) \sigma, \tag{1.3}$$

$$\implies \qquad [\sigma] = \frac{[\text{number}]}{[\text{number/area}]} = [\text{area}]. \tag{1.4}$$

This motivates the name 'cross section' and the analogy is pushed even further in the classical case: If we consider the classical scattering of a point-like particle from a hard sphere with radius R, we expect the only particles to scatter to be those that hit the sphere,

$$N = \left(\int F dt \right) \pi R^2. \tag{1.5}$$

We see that the cross section for this process has a simple interpretation as the cross-sectional area of the sphere. However, the concept of cross section can be applied in the case when the scattering process is from a target with no well defined cross-sectional area, for example a generic potential $V(r)$. In these cases, cross section retains its interpretation as a measure of probability for scattering but it is not possible to connect the cross section to a physical area in the problem, instead it is an *effective area*.

The value of the differential cross section for two colliding particles A and B depends on the reference frame. Two commonly used ones are the laboratory frame (**L**), where particle B is in the beginning at rest and centre of mass frame (**CM**). We can convert between these two frames using a

standard change of variables. One finds

$$\begin{aligned}
\tan\theta_L &= \frac{\sin\theta}{\cos\theta+\tau}\ , \\
\left(\frac{d\sigma}{d\Omega}\right)_L &= \frac{\left(1+\tau^2+2\tau\cos\theta\right)^{\frac{3}{2}}}{|1+\tau\cos\theta|}\left(\frac{d\sigma}{d\Omega}\right)_{CM}\ ,
\end{aligned}$$

with $\tau = m_A/m_B$.

In summary: the probability of a given scattering process is characterised by an effective area known as *cross section*, σ. We can define this in terms of the incoming flux, and the observed rate of particle detection at a given angle, R, by

$$\frac{d\sigma}{d\Omega} = \frac{1}{F}\frac{dR}{d\Omega}.$$

1.2 General features of potential scattering

Core reading: Chapter 13.2 of Bransden & Joachain

In this section we will consider the general description of scattering from a fixed potential $V(r)$. We will use an Ansatz for the asymptotic form for the scattering wavefunction, in terms of a quantity known as the *scattering amplitude*. We will show that this is closely related to the differential cross section.

1.2.1 The time-independent Schrödinger equation

A non-relativistic particle of mass m in a "fixed" potential, which is described by the 3-dimensional Schrödinger equation

$$i\hbar\frac{\partial}{\partial t}\Psi(\underline{r},t) = \left[-\frac{\hbar^2}{2m}\underline{\nabla}^2 + V(\underline{r})\right]\Psi(\underline{r},t)\ , \tag{1.6}$$

where $\underline{r}$ describes the distance of the particle from the origin of the potential.

As we saw in Problem 1, this equation could also describe the scattering of a particle A with mass m_A off a particle B with mass m_B due to a force which depends only on the relative distance between them,

$$i\hbar\frac{\partial}{\partial t}\Psi(\underline{r},t) = \left[-\frac{\hbar^2}{2\mu}\underline{\nabla}^2 + V(\underline{r})\right]\Psi(\underline{r},t)\,, \quad \mu := \frac{m_A m_B}{m_A + m_B}\ ,$$

where μ is the reduced mass and $\underline{r} = \underline{r}_{\mathrm{A}} - \underline{r}_{\mathrm{B}}$ is the relative distance between the two particles. This is *e.g.* the case at the LHC, where two moving protons scatter (albeit with relativistic energies). In this course we will focus on scattering from a fixed potential, but bear in mind that due to this equivalence many of our results can be interpreted more generally.

We will assume that the potential does not depend on time, and we can then separate the time dependence and we look for **stationary solutions**:

$$\Psi(\underline{r},t) =: \Psi(\underline{r})e^{-i\frac{E}{\hbar}t}\ .$$

The wave function $\Psi(\underline{r})$ has to satisfy the time-independent Schrödinger-equation

$$E\Psi(\underline{r}) = \left[-\frac{\hbar^2}{2m}\underline{\nabla}^2 + V(\underline{r})\right]\Psi(\underline{r})\ , \tag{1.7}$$

where the energy E of the particle can be written as

$$E =: \frac{\underline{p}^2}{2m} =: \frac{\hbar^2\underline{k}^2}{2m} \tag{1.8}$$

with the incident momentum of the particle $\underline{p} = \hbar \underline{k}$. We can rewrite Eq.(1.7) in a different form,

$$\left[\underline{\nabla}^2 + \underline{k}^2 - U(\underline{r})\right] \Psi(\underline{r}) = 0 \,, \tag{1.9}$$

with the rescaled potential $U(\underline{r})$ and momentum $\underline{k}$,

$$U(\underline{r}) = \frac{2m}{\hbar^2} V(\underline{r}) \,, \quad \underline{k} = \underline{p}/\hbar \,. \tag{1.10}$$

We call Eq.(1.9) the *scattering equation* and will devote several lectures to different solutions of Eq.(1.9).

1.2.2 The scattering amplitude

Now we derive a formula that directly relates the solution of the Schrödinger equation to a measurable cross section. In this subsection we do this by making several rough approximations in order to get to the desired result quickly. Later on in this lecture course we will directly derive the formula.

For large distances $|\underline{r}| \to \infty$ we split up the wave function in an incoming part and a scattered part

$$\Psi(\underline{r}) \underset{r\to\infty}{\longrightarrow} \Psi_{inc.}(\underline{r}) + \Psi_{scat.}(\underline{r}) \,.$$

For large distances we typically can neglect the potential term in Eq.(1.9) as we assume the potential has finite extent and we get the free equation

$$\left[\underline{\nabla}^2 + \underline{k}^2\right] \Psi(\underline{r}) = 0 \,, \tag{1.11}$$

Now we describe the incoming particle with a plane wave, *i.e.* all the incoming particles have the same momentum $\underline{p} = \hbar \underline{k}$ and all travel in the same direction, which we choose to be the z-direction.

$$\Psi_{inc.}(\underline{r}) = e^{i\underline{k}\cdot\underline{r}} = e^{ikz} = e^{ikr\cos\theta} \,.$$

This fulfils Eq.(1.11) as can be checked explicitly. With this normalisation we have for the particle density $\rho = |\Psi_{inc.}(\underline{r})|^2 = 1$, *i.e.* one particle per unit volume. Thus the incident flux reads

$$\begin{aligned} F &= v \cdot \rho = \frac{p}{m}\rho \,, \\ \left[\frac{1}{m^2 s}\right] &= \left[\frac{m}{s} \cdot \frac{1}{m^3}\right] \,. \end{aligned}$$

The outgoing particle we want to describe by an outgoing *radial wave*

$$\Psi_{scat.}(\underline{r}) = f(k,\theta,\phi)\frac{e^{ikr}}{r} \,,$$

where we introduce a new quantity the *scattering amplitude* $f(k,\theta,\phi)$. The factor $1/r$ is needed for this to be a solution to the free scattering equation in spherical polar coordinates. This wave function also fulfils Eq.(1.11) for very large values of r.

Thus we get for the full *asymptotic wave function*:

$$\psi_{\underline{k}} \underset{r\to\infty}{\longrightarrow} e^{i\underline{k}\cdot\underline{r}} + f(k,\theta,\phi)\frac{e^{ikr}}{r} \ . \tag{1.12}$$

This Ansatz can now be used to calculate the current density $\underline{j}$

$$\underline{j}(\underline{r}) = \frac{\hbar}{2mi}\left[\Psi^*_{scat.}(\underline{\nabla}\Psi_{scat.}) - (\underline{\nabla}\Psi^*_{scat.})\Psi_{scat.}\right] = \frac{\hbar}{m}\,\mathrm{Im}(\Psi^*_{scat.}(\underline{\nabla}\Psi_{scat.})$$

and one gets

$$j_r = \frac{\hbar k}{m}\frac{|f(k,\theta,\phi)|^2}{r^2} \ .$$

j_r represents the number of particles crossing a unit area per unit time and the detector presents a cross-sectional area $r^2 d\Omega$ to the scattered beam. The number of particles entering the detector per unit time, $d\dot{N}$ is

$$\begin{aligned} d\dot{N} &= j_r r^2 d\Omega = \frac{\hbar k}{m}\,|f(k,\theta,\phi)|^2\, d\Omega \\ &= v\,|f(k,\theta,\phi)|^2\, d\Omega \\ &= \frac{F}{\rho}\,|f(k,\theta,\phi)|^2\, d\Omega \ . \end{aligned}$$

$d\dot{N}$ can also be described by the incoming flux F and the area $d\sigma$ these particles are crossing before the scattering:

$$d\dot{N} = F d\sigma = F\frac{d\sigma}{d\Omega}d\Omega \ .$$

Equating the two expressions and using $\rho = 1$ we get our master formula

$$\frac{d\sigma}{d\Omega} = |f(k,\theta,\phi)|^2 \ . \tag{1.13}$$

If you recall our discussion of cross sections, the differential cross section is really just a normalised rate of particle detection in a given direction. This can be directly measured in an experiment, and we have found a connection between this observable quantity and the form of the solutions to the Schrödinger equation allowing us to directly test our theories.

1.3 Special functions for central potentials

Core reading: Chapter 7.3 and 7.4 of Bransden & Joachain

In this lecture, we will discuss how to compute solutions of the scattering equation. This will require a brief mathematical interlude on a class of functions known as spherical Bessel functions.

1.3.1 Separation of variables of the scattering equation

We are interested in solutions to the scattering equation,

$$\left[\nabla^2 + \underline{k}^2 - U(\underline{r})\right] \Psi_k(\underline{r}) = 0. \tag{1.14}$$

with $\underline{r} \sim (r, \theta, \phi)$. We will make a simplification and only consider *central potentials* which have no dependence on the angular variables θ and ϕ,

$$U(\underline{r}) = U(r) \ , \quad \left[U(r) = \frac{2m}{\hbar^2} V(r) \ , \ k = \frac{p}{\hbar}\right]$$

In spherical polar coordinates, we can rewrite the Laplace operator as

$$\begin{aligned} \underline{\nabla}^2 &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2}{\partial \phi^2} \\ &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) - \frac{\underline{L}^2}{\hbar^2 r^2} \ , \end{aligned}$$

with the angular momentum operator $\underline{L}$,

$$\underline{L}^2 = -\hbar^2 \left[\frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial \phi^2}\right].$$

We look to describe the most general solution as a sum of separable solutions,

$$\Psi_k(\underline{r}) \ = \ \sum_{l=0}^{\infty} \sum_{m=-l}^{+l} c_{klm} \Psi_{klm}(\underline{r}) \ , \tag{1.15}$$

where the separable solutions Ψ_{klm} are given in terms of factors with different coordinate dependence,

$$\Psi_{klm}(\underline{r}) = R_{kl}(r) Y_{lm}(\theta, \phi) \ . \tag{1.16}$$

We can find the equations which govern these factors by substitution into the Schrödinger equation. After a little algebra, we find that

$$\underline{L}^2 Y_{lm}(\theta,\phi) = C\hbar^2 Y_{lm}(\theta,\phi),$$
$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial R_{kl}}{\partial r}\right) + \left[\underline{k}^2 - U(r)\right] R_{kl}(r) = \frac{C}{r^2} R_{kl}(r).$$

The angular equation is the well-known equation for the *spherical harmonics* (see inset box) which are the simultaneous eignefunctions of the orbital momentum squared operator $\underline{L}^2$ and of the operator of the orbital momentum along the z-axis, L_z:

$$\underline{L}^2\, Y_{lm}(\theta,\phi) = \hbar^2\, l(l+1)\, Y_{lm}(\theta,\phi)\ , \tag{1.17}$$
$$L_z\, Y_{lm}(\theta,\phi) = \hbar\, m\, Y_{lm}(\theta,\phi)\ , \tag{1.18}$$

where l and m are integers such that $-l \leq m \leq l$ and $-\infty < l < +\infty$. We can therefore identify the constant as $C = l(l+1)$.

Now we consider the radial equation and expand out the derivative term,

$$\begin{aligned}
\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial R_{kl}}{\partial r}\right) + \left[\underline{k}^2 - U(r)\right] R_{kl}(r) &= \frac{l(l+1)}{r^2} R_{kl}(r), \\
\left\{\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{l(l+1)}{r^2}\right] - U(r) + \underline{k}^2\right\} R_{kl}(r) &= 0, \\
\left[\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} - U(r) + \underline{k}^2\right] R_{kl}(r) &= 0\ .
\end{aligned} \tag{1.19}$$

We will return often to the form of the radial equation in Eq.(1.19); it defines the behaviour of the radial part of the Schrödinger equation, and as we already know the angular part, to solve the scattering problem we just need to understand the behaviour of solutions to this equation.

Let's just reflect on what we've done. We have set up the time-independent Schrödinger equation for scattering solutions off a central potential. If we had a solution to this equation, we would expect the asymptotic behaviour of this solution to be of the form

$$\Psi_k(r) = R_{kl}(r) Y_{lm}(\theta,\phi) \rightarrow_{r\to\infty} e^{i\underline{k}\cdot\underline{r}} + f(k,\theta,\phi)\frac{e^{ikr}}{r}, \tag{1.20}$$

and by taking the large r limit, we could find the scattering amplitude $f(k,\theta,\phi)$, and via the arguments from Lecture 2, directly compute the scattering cross section. The only thing preventing us from doing this for an

Spherical Harmonics

The spherical harmonics can be expressed in terms of **associated Legendre functions** *and these in terms of* **Legendre polynomials** *(see your favourite textbook on mathematical physics, or google it),*

$$\begin{aligned} Y_{lm}(\theta,\phi) &= C_{lm} e^{im\phi} P_l^m(\cos\theta) \ , \\ P_l^m(x) &= (1-x^2)^{\frac{|m|}{2}} \left(\frac{d}{dx}\right)^{|m|} P_l(x) \ , \\ P_l(x) &= \frac{1}{2^l l!} \left(\frac{d}{dx}\right)^l (x^2-1)^l \ . \end{aligned}$$

The following orthogonality relations hold

$$\begin{aligned} \int Y_l^m Y_{l'}^{m'*} d\Omega &= \delta_{ll'}\delta_{mm'} \ , \\ \int_{-1}^{1} P_k^m(x) P_l^k(x) dx &= \frac{2(l+m)!}{(2l+1)(l-m)!}\delta_{ml} \ , \\ \int_{-1}^{1} P_m(x) P_n(x) dx &= \frac{2}{2n+1}\delta_{mn} \ . \end{aligned}$$

These functions form a basis for functions of (θ,ϕ) and so any function of these variables can be decomposed into spherical harmonics. If that function has additional azimuthal (cylindrical) symmetry the solutions will be even simpler and will not depend on m. In this case the spherical harmonics are simply replaced by Legendre polynomials (spherical harmonics with $m=0$):

$$Y_{lm}(\theta,\phi) \to P_l(\cos\theta) \ .$$

arbitrary potential is the radial equation in Eq. (1.19). Finding solutions to this equation is the difficult part of computing scattering amplitudes via the methods discussed so far. However, as we shall see, much can be said about the *behaviour* of these solutions even if exact expressions are rare.

1.3.2 Spherical Bessel functions

We will now look at the radial equation with no potential term

$$U(r) = 0,$$

and study the solutions. This is, of course, easier than solving the radial equation with an arbitrary potential but it is also a useful physical approximation: we will always study potentials which go to zero as r goes to infinity and the solutions in this case must approximate solutions to the zero potential radial equation. We will return to this in Lecture 4.

The simplest case: $l = 0$

To get an idea how the solutions of Eq.(1.19) might look, we start by considering the free particle case, *i.e.* $U(r) = 0$.

First we perform a useful reparameterization which is commonly used to simplify the radial derivatives. Defining $u_{kl}(r) := rR_{kl}(r)$ we are left with a simpler equation for $u_{kl}(r)$,

$$\begin{aligned} \left[\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} + \underline{k}^2\right]\frac{u_{kl}(r)}{r} &= 0 \\ \left[\frac{\partial^2}{\partial r^2} - \frac{l(l+1)}{r^2} + \underline{k}^2\right]u_{kl}(r) &= 0\ . \end{aligned} \tag{1.21}$$

Now setting $l = 0$, we get

$$\left[\frac{\partial^2}{\partial r^2} + \underline{k}^2\right]u_{k0}(r) = 0\ ,$$

which is solved by

$$u_{k0}(r) = A\sin(kr) + B\cos(kr)\ .$$

Thus we finally get

$$R_{k0} = A\frac{\sin(kr)}{r} + B\frac{\cos(kr)}{r}\ .$$

The cosine part gives a divergent contribution to R_{kl} at the origin. If we require our solution to be finite at the origin, it would imply that $B = 0$.

General case

To solve Eq. (1.21) for general values of l, we stay with the notation of the radial function R_{kl} and make a change of variables to $\rho := kr$ to get

$$\left[\frac{\partial^2}{\partial\rho^2} + \frac{2}{\rho}\frac{\partial}{\partial\rho} - \frac{l(l+1)}{\rho^2} + 1\right] R_l(\rho) = 0 \,, \tag{1.22}$$

where we denoted $R_l(\rho) := R_{kl}(\rho/k)$. This equation is known as the *spherical Bessel equation*, and methods for solving it can be found in almost any mathematical physics textbook. Here, in case this is totally new to you, we will discuss a quick way to find the solutions (although it must have taken some guess work to find).

We have actually already found the solution for $l = 0$,

$$R_0(\rho) = Ak\frac{\sin(\rho)}{\rho} + Bk\frac{\cos(\rho)}{\rho} \,.$$

Now we can derive the other solutions iteratively by using the following tricks:

- <u>Trick 1:</u> For $l \neq 0$ we make the Ansatz $R_l(\rho) =: \rho^l \chi_l(\rho)$ and Eq.(1.22) turns into
$$\left[\frac{\partial^2}{\partial\rho^2} + \frac{2(l+1)}{\rho}\frac{\partial}{\partial\rho} + 1\right] \chi_l(\rho) = 0 \,. \tag{1.23}$$
- <u>Trick 2:</u> Next we try to derive a recursion relation.
  - Assume that $\chi_l(\rho)$ is a solution of Eq.(1.23).
  - By an inspired guess, we ask what equation holds for
$$\chi(\rho) := \frac{1}{\rho}\frac{\partial\chi_l(\rho)}{\partial\rho} \;?$$

  To answer that we differentiate Eq.(1.23) once more

$$\begin{aligned}
\left[\frac{\partial^3}{\partial\rho^3} + \frac{2(l+1)}{\rho}\frac{\partial^2}{\partial\rho^2} - \frac{2(l+1)}{\rho^2}\frac{\partial}{\partial\rho} + \frac{\partial}{\partial\rho}\right] \chi_l(\rho) &= 0 \\
\left[\frac{\partial^2}{\partial\rho^2} + \frac{2(l+1)}{\rho}\frac{\partial}{\partial\rho} - \frac{2(l+1)}{\rho^2} + 1\right] (\rho\chi(\rho)) &= 0 \\
\left[\frac{\partial^2}{\partial\rho^2} + \frac{2(l+2)}{\rho}\frac{\partial}{\partial\rho} + 1\right] \chi(\rho) &= 0 \,.
\end{aligned}$$

The last line is simply the defining equation for $\chi_{l+1}(\rho)$. Thus we have the desired recursion relation

$$\chi_{l+1}(\rho) = \frac{1}{\rho}\frac{\partial \chi_l(\rho)}{\partial \rho}$$

leading to

$$\chi_l(\rho) = \left(\frac{1}{\rho}\frac{\partial}{\partial \rho}\right)^l \chi_0(\rho) \ .$$

With $\chi_0(\rho)$ given in Eq.(1.3.2) we get two fundamental solutions of Eq.(1.22): the **spherical Bessel functions** (or **Spherical Bessel functions of the first kind**) $j_l(\rho)$ and the **spherical Neumann functions** (or **spherical Bessel functions of the second kind**) $n_l(\rho)$.

$$j_l(\rho) \ := \ (-1)^l \rho^l \left(\frac{1}{\rho}\frac{\partial}{\partial \rho}\right)^l \frac{\sin\rho}{\rho} \ , \tag{1.24}$$

$$n_l(\rho) \ := \ -(-1)^l \rho^l \left(\frac{1}{\rho}\frac{\partial}{\partial \rho}\right)^l \frac{\cos\rho}{\rho} \ . \tag{1.25}$$

The factors $\pm(-1)^l$ are just convention. We call the expressions Eqs. (1.24) and (1.25) which give the spherical Bessel and Neumann functions in terms of repeated differentiation *Rodrigues' formulae*.

In summary, the solutions to the radial equation for $U(r) = 0$, Eq. (1.21), are given by linear combinations of two independent solutions $j_l(kr)$ and $n_l(kr)$

$$R_{kl}(r) = Aj_l(kr) + Bn_l(kr), \tag{1.26}$$

for complex constants A and B.

1.3.3 Properties of spherical Bessel functions

These are interesting functions, appearing in many physical applications, and listing their mathematical properties would fill (and have filled) a few learned tomes. We only need to know a little about their behaviour, which we'll highlight now, but it would be a good idea to familiarise yourself with their other properties during your wider reading.

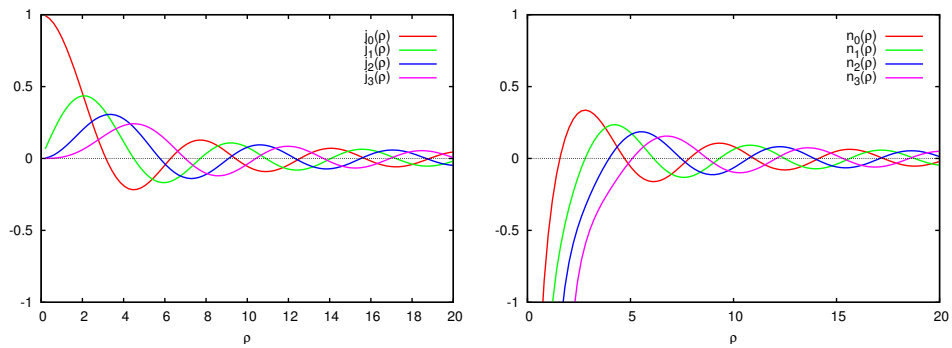

Figure 1.1: The first few spherical Bessel and Neumann functions. Note that the spherical Neumann functions diverge as we approach $\rho = 0$ whilst the spherical Bessel functions either go to one ($l = 0$) or zero ($l \neq 0$).

The first few spherical Bessel functions can be computed quickly from the Rodrigues formulae and read

$$\begin{aligned} j_0(\rho) &= \frac{\sin\rho}{\rho} \\ j_1(\rho) &= \frac{\sin\rho}{\rho^2} - \frac{\cos\rho}{\rho} \\ j_2(\rho) &= \left(\frac{3}{\rho^3} - \frac{1}{\rho}\right)\sin\rho - \frac{3}{\rho^2}\cos\rho \end{aligned} \tag{1.27}$$

and

$$\begin{aligned} n_0(\rho) &= -\frac{\cos\rho}{\rho} \\ n_1(\rho) &= -\frac{\cos\rho}{\rho^2} - \frac{\sin\rho}{\rho} \\ n_2(\rho) &= -\left(\frac{3}{\rho^3} - \frac{1}{\rho}\right)\cos\rho - \frac{3}{\rho^2}\sin\rho \end{aligned} \tag{1.28}$$

These have been plotted in Fig. 1.1, and we see that they are damped sinusoidal functions, oscillating but dying away at larger values of ρ.

The asymptotic behaviour of the spherical Bessel and Neumann functions, for both large and small values of ρ, will be particularly important for the discussion of scattering. For small values of ρ these functions can be

expanded as

$$
\begin{aligned}
j_l(\rho) &\approx \frac{2^l l!}{(2l+1)!}\rho^l \ , \\
n_l(\rho) &\approx -\frac{(2l)!}{2^l l!}\rho^{-(l+1)} \ .
\end{aligned}
$$

For large values of ρ the functions can be approximated by

$$
\begin{aligned}
j_l(\rho) &\approx \frac{1}{\rho}\sin\left(\rho - \frac{l\pi}{2}\right) \ , \\
n_l(\rho) &\approx -\frac{1}{\rho}\cos\left(\rho - \frac{l\pi}{2}\right) \ .
\end{aligned}
$$

These formula will be derived in more detail in Problems 8 and 10. Consult the solutions provided for derivations.

The spherical Bessel functions $j_l(\rho)$ and the spherical Neumann functions $n_l(\rho)$ are more or less equivalent to some other famous functions which you may come across. They get their name from their relationship to the *ordinary Bessel and Neumann functions*; the functions that we have derived can both be expressed in terms of $J_\nu(\rho)$ the ordinary Bessel function of order ν,

$$
\begin{aligned}
j_l(\rho) &= \left(\frac{\pi}{2\rho}\right)^{\frac{1}{2}} J_{l+\frac{1}{2}}(\rho) \\
n_l(\rho) &= (-1)^{l+1}\left(\frac{\pi}{2\rho}\right)^{\frac{1}{2}} J_{-l-\frac{1}{2}}(\rho)
\end{aligned}
$$

Another useful way of rewriting the spherical Bessel and Neumann functions is by taking linear combinations to make two functions which are complex conjugate to each other,

$$
\begin{aligned}
h_\nu^{(1)}(\rho) &= j_\nu(\rho) + i n_\nu(\rho) \\
h_\nu^{(2)}(\rho) &= j_\nu(\rho) - i n_\nu(\rho) = \left[h_\nu^{(1)}(\rho)\right]^*
\end{aligned}
$$

which are known as the spherical Hankel functions of the first $h_\nu^{(1)}(\rho)$ and second $h_\nu^{(2)}(\rho)$ kind. These have a simple large ρ form,

$$
h_\nu^{(1)}(\rho) = \frac{(-i)^{\nu+1}}{\rho}e^{i\rho} + \mathcal{O}\left(\frac{1}{\rho^2}\right),
$$

as can be easily checked using the asymptotic forms of $j_l(\rho)$ and $n_l(\rho)$.

1.3.4 Summary of solutions to radial equation

Assuming no potential in Eq. (1.19), the radial equation becomes

$$\left[\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r} - \frac{l(l+1)}{r^2} + \underline{k}^2\right] R_{kl}(r) = 0 \ . \tag{1.29}$$

From our discussion about the spherical Bessel equations, we have seen that this equation is generally solved by

$$R_{kl}(r) = Aj_l(kr) + Bn_l(kr) \ .$$

for two arbitrary complex constants A and B. Here k is fixed as it corresponds to the energy $E = k^2\hbar^2/(2m)$ being fixed. Therefore, the full 3-dimensional time-independent Schrödinger equation for zero potential is solved by

$$\Psi_k(\underline{r}) = \sum_{l,m} c_{klm} \left[Aj_l(kr) + Bn_l(kr)\right] Y_{lm}(\theta, \phi) \ . \tag{1.30}$$

This means that spherical harmonics in combination with spherical Bessel functions form a complete set of solutions for the free scattering equation.

We can make a simplification to the form of this solution by noting that if the solution is known to be azimuthally symmetric (independent of ϕ), only the coefficients of the $m = 0$ spherical harmonics will be non-zero. The $m = 0$ spherical harmonics are simply Legendre polynomials, so our general solution can be written as

$$\Psi_{\underline{k}}(\underline{r}) = \sum_{l=0}^{\infty} \left[A_l j_l(kr) + B_l n_l(kr)\right] P_l(\cos\theta) \ . \tag{1.31}$$

1.3.5 Plane waves

By a short calculation we can see that a plane wave of the form

$$e^{i\underline{k}\cdot\underline{r}},$$

for a fixed momentum 3-vector $\underline{k}$, is a solution to the free 3-dimensional Schrödinger equation. In fact, if we include all possible momentum 3-vectors these states form a complete set of solutions, and therefore a suitable basis for the description of the most general solution. As the plane wave is a solution, we should be able to decompose a given plane wave in terms of spherical Bessel functions and spherical Harmonics. This decomposition is

known as *Rayleigh's plane wave equation*, and for a plane wave with its momentum along the z-axis it reads

$$e^{ikz} = e^{ikr\cos\theta} \quad = \quad \sum_{l=0}^{\infty}(2l+1)i^l j_l(kr)P_l(\cos\theta).$$

In Problem 9 you will prove this formula. See the solutions for details.

Although we will not use this next result, it is nice to know that the formula above is a simple form of a more general version of Rayleigh's equation. We have a certain freedom to choose our coordinate $\underline{r}$ such that the 3-momentum $\underline{k}$ lies along the z-axis. If we then rotate back into a general coordinate basis, we find the equation

$$e^{i\underline{k}\cdot\underline{r}} = 4\pi\sum_{l=0}^{\infty}\sum_{m=-l}^{+l} i^l j_l(kr)Y^*_{lm}(\underline{k})Y_{lm}(\underline{r}) \ .$$

1.4 The method of partial waves

Core reading: Chapter 13.3 and 13.4 of Bransden & Joachain

We have seen how the scattering equation can be written in terms of spherical polar coordinates as

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \underline{k}^2 - U(r) - \frac{\underline{L}^2}{\hbar^2 r^2}\right]\Psi_{\underline{k}}(\underline{r}) = 0.$$

From this formula, it can be shown that the Hamiltonian (assuming a central potential) and the angular momentum operator squared $\underline{L}^2$ commute,

$$[H, \underline{L}^2] = 0.$$

Therefore our scattering solutions preserve the total angular momentum of the incoming wave. The initial wave in our scattering problem is a plane wave travelling along the z-axis, which is not an angular momentum eigenstate. However, the fact that H and $\underline{L}^2$ commute suggests that in some way these components of our solution are independent. The *method of partial waves* that we will study in this lecture builds on this idea by studying the behaviour of the components of the scattered waves with fixed values of $\underline{L}^2$.

1.4.1 Definition of partial waves

We start from the Ansatz for the general asymptotic scattering solution, given in Lecture 2,

$$\psi_{\underline{k}} = e^{i\underline{k}\cdot\underline{r}} + f(k,\theta,\phi)\frac{e^{ikr}}{r}\ .$$

We always consider the case when the incoming particles travel along the z-axis, so we can write the initial plane wave as

$$e^{i\underline{k}\cdot\underline{r}} = e^{ikr\cos\theta},$$

where $k = |\underline{k}|$, and the coordinate $\underline{r}$ is given in terms of spherical polar coordinates $\underline{r} \sim (r,\theta,\phi)$.

We see that the initial wave is *azimuthally symmetric* (it doesn't depend on ϕ). In fact, if we have a central potential $V = V(r)$, we expect the final solution to also have azimuthal symmetry. Recall that any function on the sphere (that is a function of θ and ϕ) can be expanded in spherical harmonics, *e.g.* for a general function $G(\theta,\phi)$ we can write

$$G(\theta,\phi) = \sum_{l=0}^{\infty}\sum_{m=-l}^{l} G_{lm}Y_{lm}(\theta,\phi),$$

with a new set of constants G_{lm}. Furthermore, if this function is azimuthally symmetric only the $m = 0$ terms can contribute, and the spherical harmonics are replaced by Legendre polynomials,

$$\text{azimuthal symmetry} \qquad \Longrightarrow \qquad G(\theta, \phi) = \sum_{l=0}^{\infty} G'_l P_l(\cos\theta),$$

for new constants G'_l. Under our assumptions, this applies to the full scattering solution $\psi_{\underline{k}}$ as well as the scattering amplitude $f(\theta, \phi)$,

$$\psi_{\underline{k}} = \sum_{l=0}^{\infty} R_l(k, r) P_l(\cos\theta) \ , \tag{1.32}$$

$$f(k, \theta, \phi) = \sum_{l=0}^{\infty} f_l(k) P_l(\cos\theta) \ . \tag{1.33}$$

Each term in the sum in Eq. (1.32) is called a **partial wave** and it is simultaneously an eigenfunction of $\underline{L}^2$ and L_z with eigenvalues $l(l+1)\hbar^2$ and 0. These different contributions are often described using spectroscopic notation: we denote the $l = 0, 1, 2, 3, , ..$ wave by the term s-wave, p-wave, d-wave, and so on.

The coefficients $f_l(k)$ in the expansion of the scattering amplitude are called **partial wave amplitudes**. According to our master formula for the cross section in terms of the scattering amplitude, we can write

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= |f(k, \theta)|^2 \\ &= \sum_{l'=0}^{\infty} \sum_{l=0}^{\infty} f^*_{l'}(k) f_l(k) P_{l'}(\cos\theta) P_l(\cos\theta) \ . \end{aligned}$$

This is greatly simplified when we integrate over the angles θ and ϕ,

$$\begin{aligned} \sigma &= \int \frac{d\sigma}{d\Omega} d\Omega = 2\pi \int_{-1}^{+1} |f(k, \theta)|^2 \, d\cos\theta \\ &= 2\pi \sum_{l'=0}^{\infty} \sum_{l=0}^{\infty} f^*_{l'}(k) f_l(k) \int_{-1}^{+1} P_{l'}(x) P_l(x) dx \\ &= 4\pi \sum_{l=0}^{\infty} \frac{1}{2l+1} |f_l(k)|^2 = \sum_{l=0}^{\infty} \sigma_l \ , \end{aligned}$$

where the second to last step uses the orthogonality of the Legendre polynomials. We see that the total cross section is a sum of contributions from each partial wave, and knowledge of the partial wave amplitudes immediately tells us the total cross section.

We can also find a relation between the partial wave amplitudes and the partial wave radial function $R_l(k,r)$ for large values of r. Recall that for all values of r, we can write the scattering solution as

$$\psi_{\underline{k}}(\underline{r}) = \sum_{l=0}^{\infty} R_l(r) P_l(\cos\theta).$$

which is just the statement that $\Psi_{\underline{k}}(\underline{r})$ is a sum of separable solutions (with azimuthal symmetry). However, at large values of r, we expect the full solution to be of the asymptotic form that we discussed in Lecture 2,

$$\psi_{\underline{k}}(\underline{r}) \to e^{i\underline{k}\cdot\underline{r}} + f(k,\theta,\phi)\frac{e^{ikr}}{r}.$$

Therefore, at large values of r, we expect the following relation

$$\begin{aligned}\sum_{l=0}^{\infty} R_l(r) P_l(\cos\theta) &= e^{i\underline{k}\cdot\underline{r}} + f(k,\theta,\phi)\frac{e^{ikr}}{r}, \\ &= \sum_{l=0}^{\infty}\left[(2l+1)i^l j_l(kr) + f_l(k)\frac{e^{ikr}}{r}\right] P_l(\cos\theta),\end{aligned}$$

where this last expression uses Rayleigh's plane wave equation and the partial wave decomposition of the scattering amplitude. Due to the orthogonality of the Legendre polynomials, this actually implies the equivalence

$$\text{(at large } r) \qquad R_l(r) = (2l+1)i^l j_l(kr) + f_l(k)\frac{e^{ikr}}{r}. \tag{1.34}$$

1.4.2 Structure of the solution and phase shifts

Next we investigate the general structure of the solution of the radial equation $R_l(k,r)$.

The most general azimuthally symmetric solution to the scattering equation can be written without approximation as

$$\psi_{\underline{k}}(\underline{r}) = \sum_{l=0}^{\infty} R_l(r) P_l(\cos\theta)$$

where the function $R_l(r)$ satisfies the radial equation,

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \underline{k}^2 - U(r) - \frac{l(l+1)}{r^2}\right] R_l(r) = 0.$$

We will always assume that the potential is only significantly non-zero near the origin. In other words, for large enough radial distances the function $U(r)$ is assumed to go to zero,

$$\lim_{r\to\infty} rU(r) = 0.$$

If we assume that $U(r) \approx 0$ for all $r \gtrsim d$ for some constant distance d (which can be seen as the range of the potential), the solutions to the radial equation should also be solutions to the radial equation *with zero potential* for values of $r \gg d$.

We have already solved the radial equation for $U(r) = 0$ and found that the solutions are, in general, linear combinations of spherical Bessel and spherical Neumann functions,

$$R_l(k, r) = B_l(k) j_l(kr) + C_l(k) n_l(kr)\ .$$

There is a subtlety here: in general, B_l and C_l should be *complex* numbers to describe the most general solution of the radial equation. However, because of a boundary condition on the solutions (we require probability to be conserved at extremely large values of r), we can actually forget that these are complex numbers and treat them as real parameters. This is what is done implicitly in the course textbook. This is a bit tricky to understand, but a proof can be found in Problems 14 and 15. If you don't like the look of those problems, feel reassured that this is extension material, you will not be penalised for assuming these constants are real. We will take the coefficients B_l and C_l to be real numbers multiplied by the same constant phase,

$$B_l = e^{i\alpha} b_l \qquad \text{and} \qquad C_l = e^{i\alpha} c_l.$$

for real b_l and c_l (the reason that this is the most general form is given in Problem 15; however, feel free to set $\alpha = 0$ and forget about it). We will see that these constants are actually all we need to know about a scattering wavefunction to compute the cross section. In a practical situation, we could determine the real constants b_l and c_l by matching with the exact solutions found in the region $r < d$, where the effects of the potential have to be taken into account.

For very large values of r we can use the asymptotic expression for the spherical Bessel and Neumann functions

$$R_l(k,r) = \frac{e^{i\alpha}}{kr}\left[b_l(k)\sin\left(kr - \frac{l\pi}{2}\right) - c_l(k)\cos\left(kr - \frac{l\pi}{2}\right)\right] . \tag{1.35}$$

This can be rewritten using a single sine function via the harmonic addition identity. We define

$$\begin{aligned} A_l(k) &:= \sqrt{b_l(k)^2 + c_l(k)^2} , \\ \tan\delta_l(k) &:= -\frac{c_l(k)}{b_l(k)} , \end{aligned}$$

which allows us to write

$$\begin{aligned} R_l(k,r) &= \frac{A_l(k)e^{i\alpha}}{kr}\left[\cos\delta_l(k)\sin\left(kr - \frac{l\pi}{2}\right) + \sin\delta_l(k)\cos\left(kr - \frac{l\pi}{2}\right)\right] \\ &= \frac{A_l(k)e^{i\alpha}}{kr}\sin\left(kr - \frac{l\pi}{2} + \delta_l(k)\right) . \end{aligned}$$

The quantities $\delta_l(k)$ are called **phase shifts** and they denote the strength of the scattering in the l-th partial wave. Computing these is the main goal of any scattering problem which is being analysed using the method of partial waves.

We can find an expression relating the partial wave scattering amplitude $f_l(k)$ and the phase shift $\delta_l(k)$. We find this by combining the asymptotic form of the radial function $R_l(r)$ with Eq. (1.34),

$$\frac{A_l(k)e^{i\alpha}}{kr}\sin\left(kr - \frac{l\pi}{2} + \delta_l(k)\right) = (2l+1)i^l\frac{\sin\left(kr - \frac{l\pi}{2}\right)}{kr} + f_l(k)\frac{e^{ikr}}{r} ,$$

which can be solved by

$$\begin{aligned} A_l(k)e^{i\alpha} &= (2l+1)i^l e^{i\delta_l(k)} , \\ f_l(k) &= \frac{2l+1}{k}e^{i\delta_l(k)}\sin\delta_l(k) . \end{aligned}$$

This final expression allows us to express the total cross section in terms of

the phase shifts,

$$f(k,\theta) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)e^{i\delta_l(k)}\sin\delta_l(k)P_l(\cos\theta)\ , \tag{1.36}$$

$$\sigma_l = \frac{4\pi}{2l+1}|f_l(k)|^2\ , \tag{1.37}$$

$$= \frac{4\pi(2l+1)}{k^2}\sin^2\delta_l(k)\ . \tag{1.38}$$

Therefore the knowledge of the scattering phase shifts is sufficient to compute the total cross section for any scattering problem,

$$\sigma = \sum_{l=0}^{\infty}\frac{4\pi}{k^2}(2l+1)\sin^2\delta_l(k).$$

This is an interesting result: we have shown that the scattering phase shifts are defined in terms of the quantities $b_l(k)$ and $c_l(k)$, and that these completely describe the cross section. These quantities must be computed by solving the full radial equation (including potential) and comparing it to the known asymptotic form at large r. It is not always easy to solve the radial equation with non-zero potential; in fact, it's usually very hard. This limits the applicability of the method of partial waves for calculations by hand. Of course, we could always numerically solve the radial equation to find the phase shifts.

A trivial example

If there is no potential at all, then the solution to the potential-less scattering problem must hold everywhere,

$$\psi_{\underline{k}}(\underline{r}) = \sum_{l=0}^{\infty}\left[A_l j_l(kr) + B_l n_l(kr)\right]P_l(\cos\theta).$$

In fact, as it must hold *everywhere*, it must hold at $r = 0$. However, the spherical Neumann functions diverge at $r = 0$, so we cannot include any of them: B_l must equal 0 for all l. Looking at our definition of $\delta_l(k)$, this means that for all l the phase shifts themselves are zero,

$$\delta_l(k) = 0.$$

Consequently, $\sigma = 0$ and there is no scattering. As expected.

1.4.3 Importance of terms in the partial wave expansion

Not all terms in the partial wave expansion are equally important, and in some cases we can safely neglect them. For large values of l the term $l(l+1)/r^2$ in the radial equation will dominate over the potential $U(r)$. Therefore we expect solution to approximate the free solution. In this case, as in our trivial example, there would be no phase shift and thus no contribution to the cross section. An estimate suggests that values of $l > ka$ can be neglected, if a is the extension of the potential.

At low energies, we can make another important approximation. In fact, for low-energy scattering this is one of the key things to know: for $k \to 0$ only the s-wave contribution will be important. This can be seen as an extension of the previous comment, but proving it rigorously is a little tricky. We will see an explicit example of the dominance of the s-wave contribution at low-energy in Problem 11.

To characterise the low-energy scattering, we can also define the **scattering length** l_s as

$$l_s := -\lim_{k\to 0} \frac{\tan\delta_0(k)}{k} \ ,$$

one finds for $k \to 0$,

$$\begin{aligned} f &\longrightarrow -l_s \ , \\ \frac{d\sigma}{d\Omega} &\longrightarrow l_s^2 \ , \\ \sigma &\longrightarrow 4\pi l_s^2 \ . \end{aligned}$$

If we compare this form to the cross section for scattering from a hard sphere of radius r_0 we see that the scattering length could be seen as an effective radius for an equivalent hard sphere scattering process. It helps to describe the strength of low-energy scattering. See Problem 11 and the discussion in the solutions about this correspondence.

1.4.4 The general solution to scattering problems

We have seen that the solution to a scattering problem is really determined by the coefficients $b_l(k)$ and $c_l(k)$ which are present in the asymptotic solution at large r,

$$R_l(k,r) = \frac{e^{i\alpha}}{kr}\left[b_l(k)\sin\left(kr - \frac{l\pi}{2}\right) - c_l(k)\cos\left(kr - \frac{l\pi}{2}\right)\right] \ . \qquad (1.39)$$

To compute these coefficients we need to be able to solve the radial equation in the region of non-zero potential. This is challenging, but we can generally imagine solving the radial equation in two parts and matching the solutions on a boundary.

In the first region, we let $r \leq d$ and the potential has its full strength. We cannot solve this in general, but let's imagine that we have found a solution $R_l^-(r)$ for $r \leq d$ which satisfies

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \underline{k}^2 - U(r) - \frac{l(l+1)}{r^2}\right] R_l^-(r) = 0.$$

We then consider the solution in the region of $r > d$. In this region, the potential is negligible, and we are solving the free radial equation for a function $R_l^+(r)$,

$$\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \underline{k}^2 - \frac{l(l+1)}{r^2}\right] R_l^+(r) = 0.$$

We know that the solution for $R_l^+(r)$ must be a combination of spherical Bessel and Neumann functions,

$$\begin{aligned} R_l^+(r) &= A_l(k)\left[b_l(k)j_l(kr) - c_l(k)n_l(kr)\right], \\ &= D_l(k)\left[j_l(kr) - \tan\delta_l(k)n_l(kr)\right]. \end{aligned}$$

with real coefficients $b_l(k)$ and $c_l(k)$ but $A_l(k)$ and $D_l(k)$ could be complex.

As always when solving the Schrödinger equation in a piece-wise manner, we must now match these two solutions at the boundary $r = d$. We require continuity and continuity of the first derivative, which provides us with two conditions

$$\begin{aligned} R_l^-(d) &= R_l^+(d), \\ \left.\frac{\partial R_l^-}{\partial r}\right|_{r=d} &= \left.\frac{\partial R_l^+}{\partial r}\right|_{r=d} \end{aligned}$$

If we know the solution $R_l^-(r)$ exactly, the only unknowns are the total normalisation $D_l(k)$ and the phase shift $\delta_l(k)$. Normally, we will want to get rid of the normalisation and find the phase shift, so we take the ratio of these constraints,

$$\frac{kj_l'(kd) - k\tan\delta_l(k)n_l'(kd)}{j_l(kd) - \tan\delta_l(k)n_l(kd)} = \left.\frac{\partial R^-}{\partial r}\right|_{r=d}\frac{1}{R^-(d)} \equiv C(k).$$

where we introduce a new function of k, $C(k)$. We can rearrange this equation to find the phase shift solely in terms of the solution inside the potential (and its derivative) on the boundary $r = d$,

$$\tan \delta_l(k) = \frac{C(k) j_l(kd) - k j_l'(kd)}{C(k) n_l(kd) - k n_l'(kd)}.$$

This is a general expression and the difficulty of applying it rests solely in finding the function $C(k)$.

The above algorithm should work for any potential which dies away as r gets large *i.e.* any potential for which we can find a distance d where the radial equation is essentially potential-less. Here are some examples of problems that you can solve:

Hard-sphere potential

This is the subject of Problem 11. See the worked solutions for this problem for a detailed discussion of how to compute the cross section using the partial wave method.

Scattering by a square well

Among the exactly solvable scattering problems via the partial wave method, this is one of the more interesting ones. You will perform the calculation in Problem 13.

An interesting feature of this problem is that the s-wave scattering phase at low energies depends parametrically on the width and depth of the well. If we describe a well of depth V_0 and width r_0, the s-wave phase shift can be written

$$k \cot \delta_0(k) = -\frac{1}{a} + \mathcal{O}(k),$$

where a is known as the *scattering length* and is given by

$$a = r_0 \left(1 - \frac{\tan\left(\sqrt{U_0} r_0\right)}{\sqrt{U_0} r_0} \right).$$

Whenever $\sqrt{U_0} r_0 = (2n+1)\pi/2$, this function diverges and $k \cot \delta_0(k)$ goes to zero. In this case, the cross section reaches a maximum. This is a form of *resonance* which will be discussed in more detail in the next section.

1.4.5 Resonances

According to Eq.(1.38) the l-th partial cross section reads

$$\sigma_l = \frac{4\pi}{k^2}(2l+1)\frac{1}{1+\cot^2\delta_l(k)} \ .$$

At some energy, we would expect $\cot\delta_l(k)$ to go to zero. This feature is is called a (l-th partial wave) *resonance*, as it leads to a local maximum of the cross section. The passing of $\cot\delta_l(k)$ through zero can happen quite quickly, and the peak associated with the maximum can be very sharp. This distinctive narrow peak is a very clear experimental signal.

The energy at which it occurs is known as the **resonance energy** E_R. At the resonance one can write

$$\cot\delta_l(k) = \frac{E_R - E}{\Gamma(E_R)/2} \ ,$$

with the **resonance width** $\Gamma(E_R)$. This leads to the **Breit-Wigner resonance formula**

$$\sigma_l = \frac{4\pi}{k^2}(2l+1)\frac{\Gamma^2(E_R)/4}{(E-E_R)^2+\Gamma^2(E_R)/4} \ ,$$

which is valid in the vicinity of a resonance. For $M \equiv E_R$, the shape of this resonance is shown in Fig. 1.2. In practice resonances are typically superimposed onto contributions from background scattering processes, *e.g.* from other partial waves. However, their distinctive shape can be quite clearly identified in many cases. In Fig. 1.3 we show an example of some resonances found at the LHC, when looking for the dimuon final state in proton-proton scattering: $p + p \to \mu^+ + \mu^-$.

1.4.6 Optical theorem

The optical theorem is a neat result relating the total scattering cross section to the forward scattering amplitude. We will prove it here using the partial wave methods, but an alternative approach will be shown in Problem 16

Recall that the partial wave scattering amplitude is given by

$$f_l(k) = \frac{2l+1}{k}e^{i\delta_l(k)}\sin\delta_l(k).$$

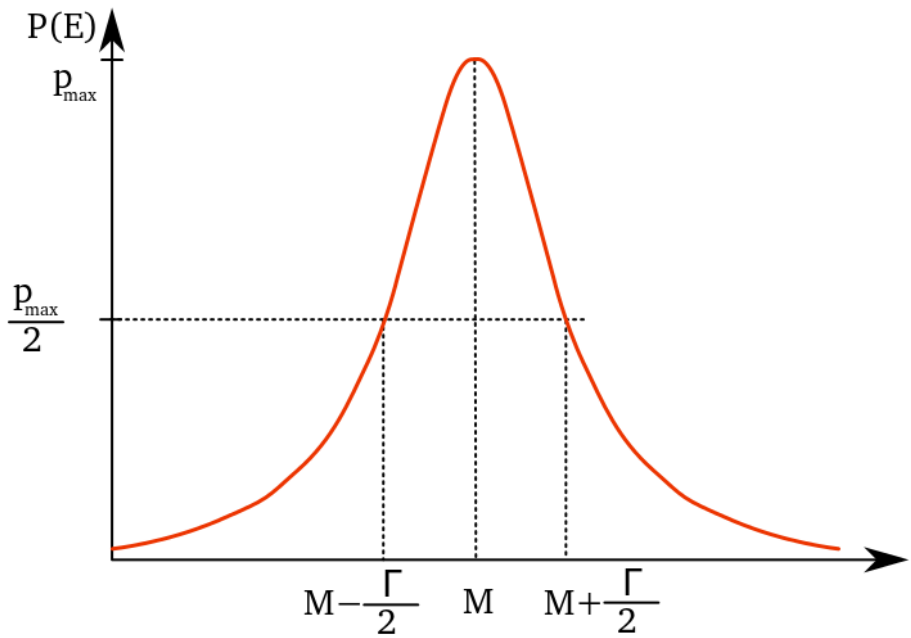

Figure 1.2: The characteristic shape of a resonance.

Therefore the full scattering amplitude can be written,

$$\begin{aligned} f(k,\theta,\phi) &= \sum_{l=0}^{\infty} f_l(k) P_l(\cos\theta), \\ &= \sum_{l=0}^{\infty} \frac{2l+1}{k} e^{i\delta_l(k)} \sin\delta_l(k) P_l(\cos\theta). \end{aligned}$$

We note that in the forward direction $\cos\theta = 1$ and the Legendre polynomials satisfy $P_l(1) = 1$ for all l. Therefore, the forward scattering amplitude is

$$f(k,0,0) = \sum_{l=0}^{\infty} \frac{2l+1}{k} e^{i\delta_l(k)} \sin\delta_l(k),$$

and its imaginary part is given by,

$$\mathrm{Im}\left[f(k,0,0)\right] = \sum_{l=0}^{\infty} \frac{2l+1}{k} \sin^2\delta_l(k).$$

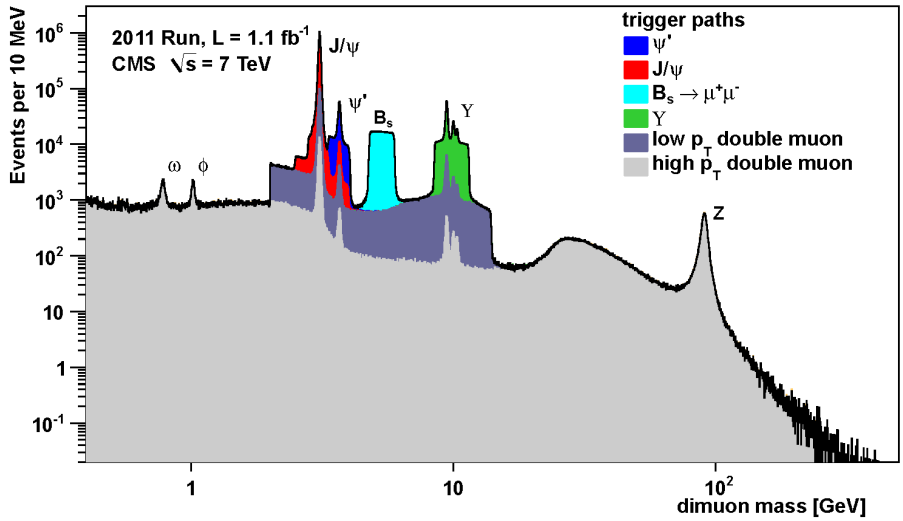

Figure 1.3: Resonances found at the LHC in the di-muon final state.

Recall that the total cross section in terms of phase shifts can be expressed as

$$\sigma = \sum_{l=0}^{\infty} \frac{4\pi}{k^2}(2l+1)\sin^2\delta_l(k), \tag{1.40}$$

$$= \frac{4\pi}{k}\mathrm{Im}\left[f(k,0,0)\right]. \tag{1.41}$$

This is the famous **Optical Theorem**: the total cross section is essentially given by the forward ($\theta = 0$) scattering amplitude. It appears in optics, quantum mechanical scattering and even quantum field theoretical scattering relevant to the LHC.

Physically it makes sense that forward scattering is related to the total rate of scattering: if there was no scattering, we would expect all particles to end up in the forward direction. Introducing a source of scattering, some of these particles must then be scattered elsewhere. Therefore if there is any scattering (the total cross section is non-zero) we must have a decrease in the relative number of particles travelling in the forward direction. One way to formalise this intuition for quantum mechanical scattering is through the conservation of the probability current. This approach will be worked out in Problem 16. See the solutions for further discussion.

1.5 The integral equation of potential scattering

Core reading: Chapter 13.5 of Bransden & Joachain

We have seen how we can solve the scattering problem via the method of partial waves. Unfortunately, this is not always easy to do: finding analytical solutions to the Schrödinger equation inside any non-trivial potential can be challenging (or impossible), and for higher energies, many partial waves could contribute to the cross section making our final expression for σ in terms of a sum of partial wave contributions impractical to perform by hand.

In this lecture we start considering non-trivial potentials and we will re-express Eq.(1.9) by an integral equation. This will allow us to reformulate the scattering problem and find expressions for the cross section without the complexities of the partial wave expansion.

1.5.1 Definition of the Green's function

We have seen before that the Schrödinger equation for the scattering problem can be recast as

$$\left[\underline{\nabla}^2 + \underline{k}^2\right] \Psi(\underline{r}) = U(\underline{r})\Psi(\underline{r}) \ . \tag{1.42}$$

The general solution of this equation can be expressed using a *Green's function*

$$\Psi(\underline{r}) = \Psi_0(\underline{r}) + \int G_0(k, \underline{r} - \underline{r}')U(\underline{r}')\Psi(\underline{r}')d\underline{r}' \ ,$$

with $\Psi_0(\underline{r})$ being a solution of the homogeneous Schrödinger equation

$$\left[\underline{\nabla}^2 + \underline{k}^2\right] \Psi_0(\underline{r}) = 0$$

and the Green's function $G_0(k, \underline{r} - \underline{r}')$ obeying the relation

$$\left[\underline{\nabla}^2 + \underline{k}^2\right] G_0(k, \underline{r} - \underline{r}') = \delta^{(3)}(\underline{r} - \underline{r}') \ . \tag{1.43}$$

If we choose the homogeneous solution to be a plane wave, we get

$$\Psi_{0,\underline{k}}(\underline{r}) = e^{i\underline{k}\cdot\underline{r}} \ , \tag{1.44}$$

$$\Psi_{\underline{k}}(\underline{r}) = e^{i\underline{k}\cdot\underline{r}} + \int G_0(k, \underline{r} - \underline{r}')U(\underline{r}')\Psi_{\underline{k}}(\underline{r}')d\underline{r}' \ . \tag{1.45}$$

1.5.2 Deriving the Green's function

In order to derive the Green's function, we consider the Fourier transformation of the definition of the Green's function, Eq.(1.43).

$$\begin{aligned}
G_0(k,\underline{r}) &= \int \frac{d^3q}{(2\pi)^3}\tilde{G}_0(k,\underline{q})e^{i\underline{q}\cdot\underline{r}} \,, \\
\left(\nabla^2+k^2\right)G_0(k,\underline{r}) &= \int \frac{d^3q}{(2\pi)^3}(k^2-q^2)\tilde{G}_0(k,\underline{q})e^{i\underline{q}\cdot\underline{r}} \stackrel{!}{=} \delta^{(3)}(\underline{r}) = \int \frac{d^3q}{(2\pi)^3}e^{i\underline{q}\cdot\underline{r}} \,, \\
\Rightarrow \tilde{G}_0(k,\underline{q}) &= \frac{1}{k^2-q^2} \,.
\end{aligned}$$

Now we have a solution for the Fourier transform of the Green's function, so we are left with transforming $\tilde{G}_0(k,\underline{q})$ back to $G_0(k,\underline{r})$.

$$\begin{aligned}
G_0(k,\underline{r}) &= \int \frac{d^3q}{(2\pi)^3}\frac{e^{i\underline{q}\cdot\underline{r}}}{k^2-q^2} \\
&\stackrel{Angular\ integr.}{=} -\frac{1}{4\pi^2 ir}\int_{-\infty}^{\infty}\frac{qe^{iqr}}{q^2-k^2}dq \\
&\stackrel{PFD}{=} -\frac{1}{4\pi r}\frac{1}{2\pi i}\int_{-\infty}^{\infty}\left(\frac{e^{iqr}}{q+k}+\frac{e^{iqr}}{q-k}\right)dq \\
&\stackrel{Cauchy}{=} -\frac{e^{ikr}}{4\pi r} \,.
\end{aligned}$$

This is the final result for the Green's function!

1.5.3 Lippmann-Schwinger equation

The general solution of the scattering equation therefore reads

$$\Psi_{\underline{k}}(\underline{r}) = e^{i\underline{k}\cdot\underline{r}} - \frac{1}{4\pi}\int \frac{e^{ik|\underline{r}-\underline{r}'|}}{|\underline{r}-\underline{r}'|}U(\underline{r}')\Psi_{\underline{k}}(\underline{r}')d\underline{r}' \,. \tag{1.46}$$

This integral equation is called the **Lippmann-Schwinger equation** and it is an equivalent reformulation of the scattering problem.
For a potential with a finite extension, *i.e.* $U(r)=0$ for $r>d$, we can investigate the limit $r\to\infty$ and $r'<d\ll r$ to get

$$\begin{aligned}
|\underline{r}-\underline{r}'| &\to r-\frac{\underline{r}}{r}\cdot\underline{r}'+\dots \,, \\
\frac{e^{ik|\underline{r}-\underline{r}'|}}{|\underline{r}-\underline{r}'|} &\to \frac{e^{ikr}}{r}e^{-i\underline{k}'\cdot\underline{r}'}+\dots \,,
\end{aligned}$$

with $\underline{k}' = \underline{r}/r\ k$. This leads to

$$\Psi_{\underline{k}}(\underline{r}) = e^{i\underline{k}\cdot\underline{r}} + \frac{e^{ikr}}{r} f(k,\theta,\phi) \ ,$$

with the scattering amplitude being given by

$$f(k,\theta,\phi) \quad = \quad -\frac{1}{4\pi}\int e^{-i\underline{k}'\cdot\underline{r}'} U(\underline{r}')\Psi_{\underline{k}}(\underline{r}')d\underline{r}.$$

We have now proven the asymptotic form of the scattering wavefunction which we assumed in Lecture 2. It follows from the asymptotic behaviour of the Green's function that the scattered wave appears radial at large r.

We can write the expression for the scattering amplitude in an alternative form by noting that it can be seen as a matrix element,

$$\begin{aligned} f(k,\theta,\phi) &= -\frac{1}{4\pi}\int e^{-i\underline{k}'\cdot\underline{r}'} U(\underline{r}')\Psi_{\underline{k}}(\underline{r}')d\underline{r}' \\ &= -\frac{1}{4\pi}\langle\Psi_{0,\underline{k}}|U|\Psi_{\underline{k}}\rangle \ . \end{aligned}$$

If we define the **transition matrix element** $T_{\underline{k}'\underline{k}}$ as

$$T_{\underline{k}'\underline{k}} := \langle\Psi_{0,\underline{k}}|V|\Psi_{\underline{k}}\rangle \ ,$$

then the differential cross section can be expressed as

$$\frac{d\sigma}{d\Omega} = \frac{m^2}{4\pi^2\hbar^4}\left|T_{\underline{k}'\underline{k}}\right|^2 \ .$$

In this language, we see that the scattering amplitude is essentially the matrix element of the potential which connects the initial state with a given outgoing state.

1.6 The Born approximation

Core reading: Chapter 13.6 of Bransden & Joachain

The Lippmann-Schwinger equation provides an alternative formulation of the scattering problem based on Green's functions. However, in its full form it is a little impractical to use as the wavefunction of interest appears on both the left hand side and inside the integral on the right hand side of the equation. This formulation does, however, suggest an iterative solution known as the *Born series*, the truncation of which provides an approximation of the scattering wavefunction (and therefore the scattering amplitude) known as the *Born approximation*.

1.6.1 The Born series

The Lippmann-Schwinger equation can be written as

$$\Psi_{\underline{k}}(\underline{r}') = e^{i\underline{k}\cdot\underline{r}'} + \int G_0(k, \underline{r}' - \underline{r}'')U(\underline{r}'')\Psi_{\underline{k}}(\underline{r}'')d\underline{r}'' \,. \tag{1.47}$$

If we replace the $\Psi_{\underline{k}}$ with the full expression given above, we find

$$\Psi_{\underline{k}}(\underline{r}) = e^{i\underline{k}\cdot\underline{r}} + \int G_0(k, \underline{r} - \underline{r}')U(\underline{r}')\left[e^{i\underline{k}\cdot\underline{r}'} + \int G_0(k, \underline{r}' - \underline{r}'')U(\underline{r}'')\Psi_{\underline{k}}(\underline{r}'')d\underline{r}''\right]d\underline{r}' \,.$$

This has produced two terms, one of which no longer depends on $\Psi_{\underline{k}}$ and the other now has two potential functions. This process of replacement can be repeated again and again. Ultimately, we get a series expansion with terms of increasing numbers of potential functions,

$$\begin{aligned}\Psi(\underline{r}) &= e^{i\underline{k}\cdot\underline{r}} + \int G_0(k, \underline{r} - \underline{r}')U(\underline{r}')e^{i\underline{k}\cdot\underline{r}'}d\underline{r}' \\ &\quad + \int G_0(k, \underline{r} - \underline{r}')U(\underline{r}')\int G_0(k, \underline{r}' - \underline{r}'')U(\underline{r}'')e^{i\underline{k}\cdot\underline{r}''}d\underline{r}''d\underline{r}' \\ &\quad + \ldots\end{aligned}$$

If the potential U is in a sense a small quantity, this series expansion can be seen as a perturbative expansion. Each term decreasing in size as its power of U increases. In this case, it would make sense to truncate the Born series after a few terms. This is called *Born approximation* and the most common version, where we keep only the terms up to first order in U, is known as the *first Born approximation*.

1.6.2 First Born approximation

The first Born approximation of the scattering wavefunction is found by truncating the Born series after the first non-trivial term (*i.e* the first term depending on the potential),

$$\Psi^1_{\underline{k}}(\underline{r}) = e^{i\underline{k}\cdot\underline{r}} + \int G_0(k, \underline{r} - \underline{r}')U(\underline{r}')e^{i\underline{k}\cdot\underline{r}'}.$$

We can see by taking the large-r asymptotic form of this expression, using the asymptotic form of the Green's function

$$G_0(k, \underline{r} - \underline{r}') \approx \left(-\frac{1}{4\pi}\right)\frac{e^{ikr}}{r}e^{-i\underline{k}'\cdot\underline{r}'},$$

and comparing this to the asymptotic form of the wavefunction that the scattering amplitude is given by

$$\begin{aligned} f^B(k,\theta,\phi) &= -\frac{1}{4\pi}\int e^{-i\underline{k}'\cdot\underline{r}}U(\underline{r})e^{i\underline{k}\cdot\underline{r}}d\underline{r} && (1.48)\\ &= -\frac{1}{4\pi}\int e^{i\underline{\Delta}\cdot\underline{r}}U(\underline{r})d\underline{r}\ , && (1.49)\end{aligned}$$

with the **momentum transfer** $\hbar\underline{\Delta}$.

$$\underline{\Delta} := \underline{k} - \underline{k}'\ . \tag{1.50}$$

Thus in the first Born approximation the scattering amplitude is simply the Fourier transform of the potential.
For elastic scattering $k = k'$ holds and thus $\underline{k}\cdot\underline{k}' = k^2\cos\theta$. Hence we get

$$\begin{aligned} \Delta^2 &= |\underline{\Delta}|^2 = 2k^2 - 2\underline{k}\cdot\underline{k}' \\ &= 2k^2(1-\cos\theta) \\ &= 4k^2\sin^2\frac{\theta}{2}\ , \\ \Rightarrow \Delta &= 2k\sin\frac{\theta}{2}\ . && (1.51)\end{aligned}$$

Therefore the Born scattering amplitude for a central potential reads

$$\begin{aligned} f^B(k,\theta,\phi) &= -\frac{1}{4\pi}\int_0^\infty drr^2U(r)\int_0^\pi d\theta'\sin\theta'\int_0^{2\pi} d\phi e^{i\Delta r\cos\theta'} \\ &= -\frac{1}{2}\int_0^\infty drr^2U(r)\int_{-1}^{+1} d\cos\theta' e^{i\Delta r\cos\theta'} \\ &= -\frac{1}{\Delta}\int_0^\infty drr\sin(\Delta r)U(r)\ . \end{aligned} \tag{1.52}$$

Important examples

The *Yukawa potential* is a particularly important example of a scattering problem that is easily treated using the Born approximation. The potential is given by

$$U_Y(r) = U_0\frac{e^{-\alpha r}}{r}\ ,$$

You will study this example in Problem 17. See the solutions for more discussion. However, the final cross section can be written as,

$$\frac{d\sigma^B}{d\Omega} = \frac{U_0^2}{(\alpha^2+\Delta^2)^2}\ .$$

Although important in its own right, the Yukawa potential actually provides a way to compute the scattering cross section for the *Coulomb potential*,

$$U_C(r) = U_0\frac{1}{r}$$

which can be seen as the $\alpha\to 0$ limit of the Yukawa potential. This potential describes the interaction for electrostatic repulsion (Coulomb's law) and is of great phenomenological importance[1]. In this case, we find the famous Rutherford scattering formula,

$$\begin{aligned} \frac{d\sigma^B}{d\Omega} &= \frac{U_0^2}{\Delta^4} \\ &= \left(\frac{q_Aq_B}{4\pi\epsilon_0}\right)^2\frac{1}{16E^2\sin^4\frac{\theta}{2}}\ . \end{aligned}$$

Other examples of potentials which can be treated via the Born approximation are given in Problems 19 and 20.

[1]For example, it is the scattering process studied by Rutherford, Geiger and Marsden.

1.7 Advanced scattering

Core reading: Chapter 13.7 and 13.8 of Bransden & Joachain

1.7.1 Collisions between identical particles

If we consider the scattering of two identical particles, then an event with the angular coordinates (θ, ϕ) cannot be distinguished from an event with the angular coordinates $(\pi - \theta, \pi + \phi)$. Thus the corresponding scattering amplitudes have to be added and *interference* can occur. The differential cross section (for finding a particle at (θ, ϕ) is given by

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= |f(\theta,\phi) + f(\pi-\theta,\pi+\phi)|^2 \\ &= |f(\theta,\phi)|^2 + |f(\pi-\theta,\pi+\phi)|^2 + 2\mathrm{Re}\left[f(\theta,\phi)f^*(\pi-\theta,\pi+\phi)\right] \ , \end{aligned}$$

where the term on the far right, which depends on both contributions to the scattering amplitude, is new to the quantum calculation. The classical expectation would see two contributions added together corresponding to the number of events scattering in process A added to the number of events scattering in process B,

$$\begin{aligned} \frac{d\sigma_{Classic}}{d\Omega} &= \frac{d\sigma_A}{d\Omega} + \frac{d\sigma_B}{d\Omega} \\ &= |f(\theta,\phi)|^2 + |f(\pi-\theta,\pi+\phi)|^2 \ . \end{aligned}$$

The additional term in the quantum case is known as an *interference* term, where due to quantum mechanics, two different processes no longer contribute independently to the total cross section. Based on the quantum statistics of the scattering particles, this will lead to significant observable differences.

Examples: spin-0 and spin-$\frac{1}{2}$

For Coulomb scattering of two spinless bosons via the Coulomb interaction, we find that

$$\begin{aligned} \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = N \Bigg\{ &\mathrm{cosec}^4\left(\frac{\theta}{2}\right) + \sec^4\left(\frac{\theta}{2}\right) \\ &+ 2\,\mathrm{cosec}^2\left(\frac{\theta}{2}\right)\sec^2\left(\frac{\theta}{2}\right)\cos\left[2\gamma\log\tan\left(\frac{\theta}{2}\right)\right]\Bigg\}, \end{aligned}$$

for an unspecified normalisation constant N (which depends on energy). This is the *Mott formula* for the scattering of two identical spinless bosons. The first term can be seen as the normal Rutherford scattering cross section, as found from our discussion of the Yukawa potential, while the second is the usual Rutherford cross section flipped by $\theta \to \pi - \theta$. The final term in this expression is due to quantum interference and leads to an oscillatory behaviour in the cross section as a function of angle around the classical cross section. See Chapter 13.7 of the core text for more details and some nice figures of this modulation in experimental data.

A similar treatment for spin-$\frac{1}{2}$ fermions, which takes into account the antisymmetry of their wavefunctions, is worked through in Chapter 13.7 of Bransden & Joachain. Please read this section, as well as Chapter 13.8, to see how we can extend our work into more realistic (and complicated!) examples involving multichannel scattering problems.

1.7.2 Form factors

Often we are scattering particles which are not elementary but are instead bound states of other particles, and accordingly have some internal properties and finite extent. We can generalise our scattering amplitude approach to take into account charge distributions via *form factors*. These will be focus on Problem 20 but we will recap some important properties here.

If we assume that the potential is given by an extended distribution of electric charge, the interaction of this distribution with another charge e we have the following potential

$$V(\underline{r}) = \int \frac{Ze^2}{s} \rho(r') d\underline{r}' \,,$$

with $\underline{s} = \underline{r} - \underline{r}'$. We can compute the first Born approximation to the scattering amplitude in the same way as before,

$$\begin{aligned}
f(k,\theta) &= -\frac{m}{2\pi\hbar^2} \int e^{i\underline{\Delta}\cdot\underline{r}} \int \frac{Ze^2}{s} \rho(r') d\underline{r}' d\underline{r} \\
&= -\frac{mZe^2}{2\pi\hbar^2} \int e^{i\underline{\Delta}\cdot(\underline{r}'+\underline{s})} \int \frac{1}{s} \rho(r') d\underline{r}' d\underline{s} \\
&= -\frac{mZe^2}{2\pi\hbar^2} \int e^{i\underline{\Delta}\cdot\underline{r}'} \rho(r') d\underline{r}' \int e^{i\underline{\Delta}\cdot\underline{s}} \frac{1}{s} d\underline{s} \\
&= -\frac{mZe^2}{2\pi\hbar^2} \widetilde{\rho}(\underline{k} - \underline{k}') \int e^{i\underline{\Delta}\cdot\underline{s}} \frac{1}{s} d\underline{s} \\
&= \tilde{\rho}(\underline{k} - \underline{k}') f_{Rutherford}(k,\theta) \,.
\end{aligned}$$

In the fourth line above, we have introduced a new quantity,

$$\widetilde{\rho}(\underline{k} - \underline{k}') = \int e^{i\underline{\Delta}\cdot\underline{r}'}\rho(r')d\underline{r}'.$$

This is just the Fourier transform of the charge distribution, and is called the **form factor** of the extended object. This is a very general result: the cross section of an extended object is given by the cross section of the corresponding point like object times the form factor squared.

$$\frac{d\sigma_{Extended}}{d\Omega} = |\tilde{\rho}|^2 \frac{d\sigma_{Point-like}}{d\Omega} \ .$$

The Fourier transformation of the charge distribution can now be obtained by comparing the measurement of the differential cross section with the calculation of the corresponding point-like cross section.

Chapter 2

Statistical Mechanics

2.1 The density matrix

Core reading: Chapter 14.1, 14.2 and 14.3 of Bransden & Joachain

2.1.1 Definition

Quantum statistical mechanics is a combination of statistical mechanics and quantum mechanics. In this case we have two sources for the necessity of a statistical description:

- A practical necessity because of a huge number of systems or degrees of freedom. Having an infinite amount of computing power, we could overcome this problem and we would not need a statistical description.

- An unavoidable necessity because of the inherent probability character of quantum mechanics. This problem cannot be overcome, even with infinite computer power.

We start our discussion of quantum statistical mechanics with some definitions:
A system that can be described by a single wave function (state vector) is said to be in a **pure state**. One implicitly assumes that a complete set of commuting observables has been determined, which is called a maximal measurement. Basically all systems we were considering till now have been in a pure state.
If in a system only non-maximal measurements are or can be made, then the wave function is not completely known and we speak of a **mixed state**. Now a statistical mixture of wave functions must be used to describe the system - this is one of the topics of quantum statistical mechanics.
Another way of stating this difference: for a pure state all sub-systems are in the same state, while for a mixed state the sub-systems can be in different states.
To make these definitions a little more clear, we consider:

- An ensemble of N sub-systems labelled by $\alpha = 1, 2, ..., N$.

- Each sub-system is in a pure state described by the state vector $\Psi^{(\alpha)}$ or $|\alpha\rangle$ in Dirac notation.
  The $|\alpha\rangle$ are normalised to one, but they are not necessarily orthogonal.

$$\langle\alpha|\alpha\rangle = 1 \ . \tag{2.1}$$

- Choose a complete set of basis vectors $|n\rangle$, i.e. orthogonal eigenvectors of a complete set of operators:

$$\langle n|n'\rangle = \delta_{n,n'} \tag{2.2}$$

or

$$\langle n|n'\rangle = \delta(n-n') \ , \tag{2.3}$$

if n is a continuous variable.
Completeness tells us that we can write the identity as

$$1 = \sum_n |n\rangle\langle n| \ . \tag{2.4}$$

The pure state $|\alpha\rangle$ can be expanded in terms of the basis $|n\rangle$:

$$|\alpha\rangle = \sum_n c_n^{(\alpha)}|n\rangle \ ,$$

with

$$\begin{aligned} c_n^{(\alpha)} &= \langle n|\alpha\rangle \ , \\ \sum_n \left|c_n^{(\alpha)}\right|^2 &= 1 \ . \end{aligned}$$

Next we consider an observable represented by an operator A. The expectation value of A in the state $|\alpha\rangle$ is

$$\begin{aligned} \langle A\rangle_\alpha &= \langle\alpha|A|\alpha\rangle \\ &= \sum_n\sum_{n'} c_{n'}^{(\alpha)*}c_n^{(\alpha)}\langle n'|A|n\rangle \\ &= \sum_n\sum_{n'}\langle n|\alpha\rangle\langle\alpha|n'\rangle\langle n'|A|n\rangle \ . \end{aligned}$$

The average of A over the ensemble - the **ensemble** or **statistical average** is denoted by

$$\langle A\rangle = \sum_{\alpha=1}^{N} W_\alpha\langle A\rangle_\alpha \ , \tag{2.5}$$

with W_α being the **statistical weight** of the pure state $|\alpha\rangle$, i.e. the probability of finding the whole system in the state $|\alpha\rangle$:

$$0 \le W_\alpha \le 1 \ , \tag{2.6}$$

$$\sum_{\alpha=1}^{N} W_\alpha = 1 \ . \tag{2.7}$$

All in all we can now write

$$\begin{aligned}
\langle A \rangle &= \sum_{\alpha=1}^{N} \sum_{n} \sum_{n'} \langle n|\alpha\rangle W_\alpha \langle \alpha|n'\rangle \langle n'|A|n\rangle \\
&= \sum_{n} \sum_{n'} \langle n|\rho|n'\rangle \langle n'|A|n\rangle \\
&= \sum_{n} \langle n|\rho A|n\rangle \\
&= \mathrm{Tr}(\rho \mathrm{A}) \ , \qquad (2.8)
\end{aligned}$$

with the **density operator** or **statistical operator**, being defined as

$$\rho = \sum_{\alpha=1}^{N} |\alpha\rangle W_\alpha \langle \alpha| \ . \qquad (2.9)$$

<u>Remarks:</u>

- The knowledge of the density operator enables us to calculate ensemble averages.
- Taking A to be the unit operator and keeping in mind that the pure states are normalised to one, we have

  $$\mathrm{Tr}(\rho) = 1 \ .$$

  If the states $|\alpha\rangle$ are not normalised to one, we get

  $$\langle A \rangle = \frac{\mathrm{Tr}(\mathrm{A}\rho)}{\mathrm{Tr}(\rho)} \ .$$

- The density operator can also be expressed in terms of matrix elements - the **density matrix**.
  - <u>Definition:</u>

$$\begin{aligned}
\rho_{nn'} &= \langle n|\rho|n'\rangle \\
&= \sum_{\alpha=1}^{N} \langle n|\alpha\rangle W_\alpha \langle \alpha|n'\rangle \\
&= \sum_{\alpha=1}^{N} W_\alpha c_{n'}^{(\alpha)*} c_n^{(\alpha)} \ .
\end{aligned}$$

– Diagonal elements: the diagonal elements of the density matrix

$$\rho_{nn} = \langle n|\rho|n\rangle = \sum_{\alpha=1}^{N} W_\alpha \left|c_n^{(\alpha)}\right|^2 \tag{2.10}$$

have the following interpretation:
W_α is the probability of finding the whole system in the state $|\alpha\rangle$ and $\left|c_n^{(\alpha)}\right|^2$ is the probability of finding that $|\alpha\rangle$ to be in the state $|n\rangle$. Thus ρ_{nn} gives the probability of finding the ensemble in the state $|n\rangle$.
Moreover we have

$$0 \leq \rho_{nn} \leq 1 \ . \tag{2.11}$$

The l.h.s. follows from Eq.(2.10) and the r.h.s. from Eq.(2.1.1).

– Hermiticity: the density matrix is hermitian

$$\langle n|\rho|n'\rangle = \langle n'|\rho|n\rangle^* \ .$$

Thus it can be diagonalised by a unitary transformation. If we choose a representation $|k\rangle$ where ρ is diagonal, then we have

$$\rho_{kk'} = \rho_{kk}\delta_{kk'} \ . \tag{2.12}$$

In that basis we immediately see from Eq.(2.11)

$$\mathrm{Tr}(\rho^2) \leq \mathrm{Tr}(\rho) = 1 \ . \tag{2.13}$$

This relation is also valid in any other representation, since the trace is invariant under unitary transformations. Thus we have

$$\sum_n \sum_{n'} |\rho_{nn'}|^2 \leq 1 \ .$$

Now let us assume that the whole system is in a pure state $|\lambda\rangle$, meaning $W_\alpha = \delta_{\alpha\lambda}$. So the density operator reads

$$\rho^\lambda = |\lambda\rangle\langle\lambda| \ ,$$

which is simply a projection operator with

$$(\rho^\lambda)^2 = \rho^\lambda \ .$$

Hence Eq.(2.13) becomes for a pure state

$$\mathrm{Tr}\left[(\rho^\lambda)^2\right] = \mathrm{Tr}\left[\rho^\lambda\right] = 1 \ .$$

This is now a new criterion for deciding whether a state was pure or not!

$$\mathrm{Tr}\left[\rho^2\right] = \ 1 \ \Rightarrow \text{state is pure} \ , \tag{2.14}$$
$$\mathrm{Tr}\left[\rho^2\right] < \ 1 \ \Rightarrow \text{state is mixed} \ . \tag{2.15}$$

If the system is in a pure state we have also

$$\langle A \rangle = \langle \lambda | A | \lambda \rangle \ .$$

If we use a diagonal representation in that case we get in addition

$$\rho^\lambda_{kk'} = \delta_{k\lambda}\delta_{k'\lambda} \ .$$

Thus one diagonal element is one and the rest zero.

– The indices n, n' denote all possible quantum numbers, if we are e.g. only interested in spin, we have a **reduced density matrix**.

2.1.2 Von Neumann equation

The density matrix provides a way to describe all the information about the state of a quantum system, including statistical randomness. We need to be able to evolve this state in time, to see how our system changes under the influence of the Hamiltonian. This evolution is described by the *von Neumman* equation, which we will now derive.

In the Schrödinger-picture we have at the time t_0

$$\rho(t_0) = \sum_{\alpha=1}^{N} W_\alpha |\alpha(t_0)\rangle\langle\alpha(t_0)| \ .$$

The time evolution of the states is given as

$$|\alpha(t)\rangle = U(t, t_0)|\alpha(t_0)\rangle \ ,$$

with

$$U(t, t_0) = e^{-i\frac{H}{\hbar}(t-t_0)}$$

for a time-independent Hamiltonian. Thus the density operator at time t is given as

$$\begin{aligned}\rho(t) &= \sum_{\alpha=1}^{N} W_\alpha |\alpha(t)\rangle\langle\alpha(t)| \\ &= U(t,t_0)\rho(t_0)U^\dagger(t,t_0) \\ &= e^{-i\frac{H}{\hbar}(t-t_0)}\rho(t_0)e^{+i\frac{H}{\hbar}(t-t_0)} \ .\end{aligned}$$

This last equation is equivalent to

$$i\hbar\frac{\partial}{\partial t}\rho(t) = [H,\rho(t)] \ , \tag{2.16}$$

which is called the **von-Neumann-equation** or sometimes also Liouville-von-Neumann-equation.

<u>Remarks:</u>

- Eq.(2.16) is the equivalent of the **Liouville-equation** for the phase space probability in classical mechanics.

$$\frac{d}{dt}\rho = \{H,\rho\} \ .$$

- Eq.(2.16) looks also similar to the **Heisenberg equation** of motion for an observable,

$$i\hbar\frac{d}{dt}A_H = -[H,A_H] + i\hbar\frac{\partial}{\partial t}A_H \ ,$$

  but keep in mind that in Eq.(2.16) the states are given in the Schrödinger-picture.

- The time evolution of the statistical average of an operator A that is not explicitly time-dependent reads

$$\begin{aligned}\frac{\partial}{\partial t}\langle A\rangle &= \frac{\partial}{\partial t}\mathrm{Tr}\,(\rho A) \\ &= \mathrm{Tr}\left(A\frac{\partial}{\partial t}\rho\right) \\ &= -\frac{i}{\hbar}\mathrm{Tr}\,(A\,[H,\rho]) \\ &= -\frac{i}{\hbar}\mathrm{Tr}\,(AH\rho - A\rho H) \\ &= -\frac{i}{\hbar}\mathrm{Tr}\,([A,H]\,\rho) \ .\end{aligned}$$

2.2 Quantum mechanical ensembles

Core reading: Chapter 14.4 and 14.5 of Bransden & Joachain

In the following we will discuss different classes of ensembles. We consider only systems in thermal equilibrium, thus the averages of physical observables are constant. In that case we get a time-independent density matrix

$$\frac{\partial}{\partial t}\rho = 0 \ ,$$

which is equivalent to H having no explicit time dependence and

$$[H, \rho] = 0 \ .$$

Now we can choose a basis $|n\rangle$ in which both H and ρ are diagonal

$$\begin{aligned} \langle n|H|m\rangle &= H_{nm} = E_n \delta_{nm} \ , \\ \langle n|\rho|m\rangle &= \rho_{nm} = \rho_{nn}\delta_{nm} \ . \end{aligned}$$

The probability that a sub-system chosen randomly from the ensemble is in the eigenstate $|n\rangle$ is ρ_{nn}. Since ρ can be expressed in terms of H

$$\rho = \rho(H) \ ,$$

we can also express ρ_{nn} in terms of E_N.
Below we present different functional dependencies of ρ_{nn} from E_n, which define different ensembles.

2.2.1 Micro-canonical ensemble

A micro-canonical ensemble describes a closed system, i.e. we have for each sub-system:

- a fixed number of particles $\mathcal{N}$;
- a fixed volume $\mathcal{V}$;
- the energy E lies in the small interval

$$E_0 \leq E \leq E_0 + \Delta \ .$$

<u>Notation:</u>

- Γ: the total number of distinct pure states in which a member of the ensemble can be found

$$\Gamma = \Gamma(\mathcal{N}, \mathcal{V}, E_0, \Delta) \ .$$

- $\mathcal{H}$: the Hamiltonian of each sub-system.

Now we make the **fundamental postulates:**

1. There is an equal probability of finding the system in any of these states, i.e.

$$\rho_{nn} = \begin{cases} 1/\Gamma & \text{for } \ E_0 \leq E \leq E_0 + \Delta \\ 0 & \text{else} \end{cases} \ .$$

2. The members of the ensemble do not interact, so the corresponding wave functions do not interfere. This can be obtained by demanding, that the wave functions of each member $|\alpha\rangle$ of the ensemble are distributed at random or equally: the phases of the probability amplitudes $c_n^{(\alpha)}$ are distributed at random.

The **entropy** of the system, is defined as

$$S = k_B \log \Gamma \ ,$$

with the Boltzmann constant $k_B = 1.3806488(13) \cdot 10^{-23}\text{J K}^{-1}$.

<u>Remarks:</u>

- The definition agrees with the usual thermodynamical one, but now Γ is calculated within using quantum mechanics.
- If each sub-system is in the same state we have $\Gamma = 1$ and thus $S = 0$ and the system is perfectly ordered.
- If Γ is greater than 1, the system is disordered. Thus S is a measure of disorder.

The entropy has the familiar properties:

1. The entropy of a closed system is maximal, when the system is in equilibrium.
2. The entropy is additive, i.e. if a closed system is divided into parts with the entropy S_1 and S_2 we have $S = S_1 + S_2$.

3. The temperature T of the system is defined as

$$\frac{1}{T} = \frac{\partial S}{\partial E} \ .$$

In practice the micro-canonical ensemble is difficult to use, thus Gibbs introduced two additional concepts: the canonical ensemble and the grand canonical ensemble.

2.2.2 Canonical ensemble

Definition:

- In this case the investigated system $\mathcal{S}$ is in thermal contact with a much larger heat reservoir $\mathcal{R}$. The combined system $\mathcal{S} + \mathcal{R}$ is closed and its statistical properties are defined by the micro-canonical ensemble.
- The system $\mathcal{S}$ is composed of a fixed number of particles $\mathcal{N}_{\mathcal{S}}$ within a volume $\mathcal{V}_{\mathcal{S}}$ and it is in the energy eigenstate $E_{\mathcal{S}}$. The system $\mathcal{R}$ is composed of a fixed number of particles $\mathcal{N}_{\mathcal{R}}$ within a volume $\mathcal{V}_{\mathcal{R}}$ and it is in the energy eigenstate $E_{\mathcal{R}}$.
- The systems $\mathcal{S}$ and $\mathcal{R}$ are in contact and can thus exchange energy - the interaction is supposed to be so weak, that at any time $\mathcal{S}$ and $\mathcal{R}$ are in definite energy eigenstates with energies $E_{\mathcal{R}}$ and $E_{\mathcal{S}}$.
- In equilibrium $\mathcal{S}$ and $\mathcal{R}$ have a common temperature T.
- Conservation of energy tells us

  $$E = E_{\mathcal{R}} + E_{\mathcal{S}} \ ,$$

  where E is an eigenvalue of the combined system. Since the whole system is described by the micro-canonical ensemble we have

  $$E_0 \leq E \leq E_0 + \Delta \ \text{ with } \ \Delta \ll E_0 \ .$$

Problem: Determine the probability that at a given temperature T the system $\mathcal{S}$ is to be found in a state with energy $E_{\mathcal{S}}$.

Solution:

- $d\Gamma(E)$: total number of distinct states of the combined system with energies in $[E, E + dE]$.

- The probability of finding the combined system with energies in $[E, E+dE]$ is

$$dP = \begin{cases} Cd\Gamma(E) & E_0 \leq E \leq E_0 + \Delta \\ 0 & \text{else} \end{cases} ,$$

where C is a constant describing the probability of finding one state.

- Denoting the corresponding numbers in the system $\mathcal{S}$ and $\mathcal{R}$ by $d\Gamma_{\mathcal{S}}$ and $d\Gamma_{\mathcal{R}}$, we have

$$\begin{aligned} d\Gamma &= d\Gamma_{\mathcal{S}} d\Gamma_{\mathcal{R}} , \\ dP &= Cd\Gamma_{\mathcal{S}}(E_{\mathcal{S}}) d\Gamma_{\mathcal{R}}(E_{\mathcal{R}}) \delta(E - E_{\mathcal{S}} - E_{\mathcal{R}}) , \end{aligned}$$

where the δ-function expresses the energy conservation.

- dP_n is the probability that the system $\mathcal{S}$ has the energy E_n without regard to the state of the reservoir (i.e. we are summing over all the states in the reservoir that yield the same E_n)

$$\begin{aligned} dP_n &= Cd\Gamma_{\mathcal{S}}(E_n) \int \delta(E - E_n - E_{\mathcal{R}}) d\Gamma_{\mathcal{R}}(E_{\mathcal{R}}) \\ &= Cd\Gamma_{\mathcal{S}}(E_n) \Delta\Gamma_{\mathcal{R}}(E - E_n) , \end{aligned}$$

with $\Delta\Gamma_{\mathcal{R}}(E - E_n)$ being the total number of states in the reservoir, subject to the conservation of energy.

- The corresponding entropy of the reservoir is given as

$$\begin{aligned} S_{\mathcal{R}} &= k_B \log \Delta\Gamma_{\mathcal{R}} , \\ \Delta\Gamma_{\mathcal{R}} &= e^{\frac{S_{\mathcal{R}}}{k_B}} . \end{aligned}$$

- Since the system $\mathcal{S}$ is supposed to be very small compared to the total system, we have $E_n \ll E$ and thus we can Taylor expand:

$$\begin{aligned} S_{\mathcal{R}}(E_{\mathcal{R}}) &= S_{\mathcal{R}}(E - E_n) \\ &= S_{\mathcal{R}}(E) - \frac{\partial S_{\mathcal{R}}(E)}{\partial E} E_n + ... \\ &= S_{\mathcal{R}}(E) - \frac{E_n}{T} + \dots . \end{aligned}$$

- Putting now the pieces together we get

$$\begin{aligned} dP_n &= C d\Gamma_{\mathcal{S}}(E_n)\Delta\Gamma_{\mathcal{R}}(E-E_n) \\ &= C d\Gamma_{\mathcal{S}}(E_n) e^{\frac{S_{\mathcal{R}}(E-E_n)}{k_B}} \\ &= C d\Gamma_{\mathcal{S}}(E_n) e^{\frac{S_{\mathcal{R}}(E)}{k_B}} e^{-\frac{E_n}{k_B T}} \\ &= \mathcal{A} e^{-\frac{E_n}{k_B T}} d\Gamma_{\mathcal{S}}(E_n) \ . \end{aligned}$$

  This is already the desired result: the probability of finding the system $\mathcal{S}$ in the energy state with E_n is given by $\mathcal{A}e^{-\frac{E_n}{k_B T}}$.

- Thus we can write the density matrix as

$$\rho_{nm} = \mathcal{A} e^{-\beta E_n} \delta_{nm} \ ,$$

  with $\beta = 1/(k_B T)$.

<u>Remarks:</u>

1. The coefficient $\mathcal{A}$ can be extracted from the condition $\mathrm{Tr}(\rho) = 1$.

$$\begin{aligned} 1 &= \sum_n \rho_{nn} \\ &= \sum_n \mathcal{A} e^{-\beta E_n} \\ \Rightarrow \mathcal{Q}_{\mathcal{N}_{\mathcal{S}}} := \mathcal{A}^{-1} &= \sum_n e^{-\beta E_n} = \mathrm{Tr}\left[e^{-\beta H}\right] . \end{aligned} \tag{2.17}$$

   $\mathcal{Q}_{\mathcal{N}_{\mathcal{S}}}$ is called the **partition function**.

2. The sum runs over all distinct states, which is not necessarily the same as over all energy eigenstates.

3. The density operator can now be written as

$$\rho = \frac{e^{-\beta H}}{\mathcal{Q}_{\mathcal{N}_{\mathcal{S}}}} \ .$$

4. The ensemble average of an operator A is now given as

$$\begin{aligned} \langle A \rangle &= \mathrm{Tr}(\rho A) \\ &= \frac{1}{\mathcal{Q}_{\mathcal{N}_{\mathcal{S}}}} \mathrm{Tr}\left(e^{-\beta H} A\right) \\ &= \frac{\mathrm{Tr}\left(e^{-\beta H} A\right)}{\mathrm{Tr}\left[e^{-\beta H}\right]} \ . \end{aligned}$$

5. The average energy $\bar{E}$ of the system can be found by setting $A = H$. Using Eq.(2.17) it can also be expressed as

$$\bar{E} = \langle H \rangle = -\frac{1}{\mathcal{Q}_{\mathcal{N}_\mathcal{S}}} \frac{\partial \mathcal{Q}_{\mathcal{N}_\mathcal{S}}}{\partial \beta} = -\frac{\partial}{\partial \beta} \log \mathcal{Q}_{\mathcal{N}_\mathcal{S}} \ .$$

2.2.3 Grand canonical Ensemble

This is similar the canonical ensemble, but now we allow in addition the exchange of particles between the system $\mathcal{S}$ and the reservoir $\mathcal{R}$. The total number of particles

$$\mathcal{N} = \mathcal{N}_\mathcal{R} + \mathcal{N}_\mathcal{S}$$

is constant.
All derivations are similar up to difference in the Taylor expansion of the entropy, where now the change of particle numbers is taken into account.

$$\begin{aligned} \mathcal{S}_\mathcal{R}(E - E_n, \mathcal{N} - \mathcal{N}_\mathcal{S}) &= \mathcal{S}_\mathcal{R}(E, \mathcal{N}) \\ &\quad - \frac{\partial \mathcal{S}_\mathcal{R}(E,\mathcal{N})}{\partial E} E_n(\mathcal{N}_\mathcal{S}) - \frac{\partial \mathcal{S}_\mathcal{R}(E,\mathcal{N})}{\partial \mathcal{N}} \mathcal{N}_\mathcal{S} + \dots \ . \\ &= \mathcal{S}_\mathcal{R}(E,\mathcal{N}) - \frac{E_n(\mathcal{N}_\mathcal{S})}{T} + \frac{\mu}{T} \mathcal{N}_\mathcal{S} + \dots \ . \end{aligned} \tag{2.18}$$

The density matrix gets now the form

$$\rho_{nm} = \mathcal{B} e^{[-\beta E_n(\mathcal{N}_\mathcal{S}) - \alpha \mathcal{N}_\mathcal{S}]} \delta_{nm} \ ,$$

with $\mathcal{B}$ being a constant and $\alpha = (\partial \mathcal{S}_\mathcal{R} / \partial \mathcal{N} / k_B)$. $\mu = -\alpha/\beta$ is called **chemical potential**.
In order to give an operator form of the density operator, we introduce the number operator $\nu_\mathcal{S}$, that is counting the number of particles in the system $\mathcal{S}$. Its eigenvalues are $N = 0, 1, 2, \dots$. Thus we get for the density operator

$$\rho = \mathcal{B} e^{-\beta H(\nu_\mathcal{S}) - \alpha \nu_\mathcal{S}} \ .$$

The constant $\mathcal{B}$ can again be determined from $\mathrm{Tr}(\rho) = 1$ and we get

$$\mathcal{B}^{-1} := Z(\alpha, T) = \mathrm{Tr}\left[e^{-\beta H(\nu_\mathcal{S}) - \alpha \nu_\mathcal{S}} \right] \ .$$

Z is called the **grand partition function** and we can write

$$\begin{aligned} Z(\alpha, T) &= \sum_N \sum_n e^{-\beta E_n(N) - \alpha N} \\ &= \sum_N \left[\sum_n e^{-\beta E_n(N)} \right] e^{-\alpha N} \\ &= \sum_N \left[\sum_n e^{-\beta E_n(N)} \right] \left[e^{-\alpha} \right]^N \\ &= \sum_N \mathcal{Q}_N(T) z^N \ , \end{aligned}$$

where $z = exp(-\alpha)$ is called **fugacity** and $\mathcal{Q}_N(T)$ is the partition function of a system of N particles.

2.2.4 Applications

See section 14.5 of Bransden & Joachain for a discussion of how to apply the canonical ensemble formalism discussed in this lecture to a spin-$\frac{1}{2}$ particle in a magnetic field and to a particle in a box.

2.3 Systems of non-interacting particles

Core reading: Chapter 14.6 of Bransden & Joachain

We will now discuss systems of large numbers of non-interacting objects (particles) which are equivalent and possess the same energy levels.
$E_j (j = 1, 2, 3, ..)$ denotes an energy level of one of the particles. The total energy E of the system can be written as

$$E_{\mathcal{S}} = E = \sum_j n_j E_j \ ,$$

where the sum runs over all energy eigenstates and n_j is the number of particles having the same energy E_j. The total number of particles in the system is

$$N_{\mathcal{S}} = N = \sum_j n_j \ .$$

The individual particle numbers can have the following values:

$$n_j = \begin{cases} 0, 1, 2, 3, ... & \text{Maxwell-Boltzmann} & \text{particles distinguishable} \\ 0, 1, 2, 3, ... & \text{Bose-Einstein} & \text{particles indistinguishable - bosons} \\ 0, 1 & \text{Fermi-Dirac} & \text{particles indistinguishable - fermions} \end{cases} .$$

For obtaining the partition functions we have to sum over all distinct states and not only over all energy eigenstates. Thus we need to know the number of distinct states g_E having the same energy E.
To find that we start by looking at the total wave function of a system and we find

$$\Psi = \begin{cases} \Phi_{E_j}(1)\Phi_{E_k}(2) \cdot \cdots \cdot \Phi_{E_l}(N) & \text{Maxwell-Boltzmann} \\ \text{total symmetric} & \text{Bose-Einstein} \\ \text{total antisymmetric} & \text{Fermi-Dirac} \end{cases} .$$

This leads to

$$g_E = \begin{cases} g_E^{MB} = \frac{N!}{\Pi_j (n_j!)} & \text{Maxwell-Boltzmann} \\ g_E^{BE} = 1 & \text{Bose-Einstein} \\ g_E^{FD} = 0, 1 & \text{Fermi-Dirac} \end{cases} .$$

<u>Remarks:</u>

- ad MB: If all energies $E_j, E_k, ..., E_l$ are different, then each energy level is $N!$-fold degenerate. If, however, n_j particles with $n_j > 1$ have the same energy, then an interchange among them does not change the wave function.

- ad BE: there is only one total symmetric wave function.
- ad FD: there is only one total anti-symmetric wave function. If $n_j > 1$ then g_E becomes zero.

Now the partition function reads for all the three cases:

$$Q_N(T) = \sum_{\{n_j\}} g_E \exp\left[-\beta \sum_j n_j E_j\right] ,$$

where $\sum\limits_{\{n_j\}}$ denotes a sum over every possible set of numbers n_j that is compatible with the total number N.
The average value of the energy $\langle E \rangle$ of the system can be calculated if the average value $\langle n_i \rangle$ of each occupation number n_i is known.

$$\langle E \rangle = \sum_i E_i \langle n_i \rangle .$$

$\langle n_i \rangle$ can be determined from the canonical ensemble

$$\langle n_i \rangle = \frac{1}{Q_N(T)} \sum_{\{n_j\}} g_E \exp\left[-\beta \sum_j n_j E_j\right] n_i ,$$

which is equivalent to

$$\langle n_i \rangle = -\frac{\partial \log Q_N(T)}{\partial(\beta E_i)} .$$

For the case of MB the sum can be performed explicitly to obtain

$$Q_N(T) = \left[\sum_j \exp\left(-\beta E_j\right)\right]^N ,$$

leading to

$$\langle n_i \rangle = \exp\left[-\left(\beta E_i + \alpha\right)\right] ,$$

with

$$\alpha = \log\left[\frac{\sum\limits_j \exp\left(-\beta E_j\right)}{N}\right] .$$

For the case of BE and FD it is easier to start with the grand partition function

$$\begin{aligned} Z &= \sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\ldots g_E \exp\left\{-\beta\sum_j\left[n_jE_j(N)\right]-\alpha\sum_j n_j\right\} \\ &= \sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\ldots g_E \prod_j \exp\left\{-n_j\left[\beta E_j(N)+\alpha\right]\right\} \\ &= \prod_j\sum_{n_j=0}^{\infty} g_E \exp\left\{-n_j\left[\beta E_j(N)+\alpha\right]\right\} \ . \end{aligned}$$

In the case of BE we have $g_E = 1$ and n_j has values from 0 to ∞. Thus we have a simple geometric series and get

$$Z^{BE} = \prod_j \frac{1}{1-\exp\left\{-\left[\beta E_j(N)+\alpha\right]\right\}} \ .$$

In the case of FD we have $g_E = 0, 1$ and n_j has only the values 0 and 1. So we simply can write down the two terms and get

$$Z^{FD} = \prod_j \left[1+\exp\left\{-\left[\beta E_j(N)+\alpha\right]\right\}\right] \ .$$

The average number of states is now given as

$$\begin{aligned} \langle n_i\rangle &= \frac{1}{Z}\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\ldots g_E \exp\left\{-\beta\sum_j\left[n_jE_j(N)\right]-\alpha\sum_j n_j\right\} n_i \\ &= -\frac{\partial \log Z}{\partial\left[\beta E_i(N)\right]} \ . \end{aligned}$$

Now all three cases (MB, BE and FD) can be expressed with one formula:

$$\langle n_i\rangle = \frac{1}{\exp\left[\beta E_i(N)+\alpha\right]+\gamma}$$

with

$$\gamma = \begin{cases} 0 & \text{MB} \\ -1 & \text{BE} \\ +1 & \text{FD} \end{cases} \ .$$

The function

$$\langle n(E)\rangle = \frac{1}{\exp\left[\beta E+\alpha\right]+\gamma}$$

is called the **distribution function**.

2.3.1 Behaviour of the distribution function

The distribution function has a simple form for all three types of particles. At very high values of the argument, $\beta E+\alpha$, we find that all types of particle have a similar distribution,

$$n(E) = \frac{1}{e^{\beta E+\alpha} + \gamma} \approx \frac{1}{e^{\beta E+\alpha}}.$$

This is just a falling exponential.

The behaviour is more interesting when $\beta E + \alpha$ gets small. For indistinguishable bosons, we find that the denominator can get arbitrarily small, the exponential cancelling with the $\gamma = -1$, leading to very large values of the distribution function. This corresponds to *Bose-Einstein condensation* where we get a macroscopic number of particles occupying the lowest energy level. The picture is entirely different for fermions. If we use the chemical potential μ, such that $\alpha = -\beta\mu$, the distribution function can be written

$$n(E) = \frac{1}{e^{\beta(E-\mu)} + 1}.$$

For very large values of β (very low temperatures), we see qualitatively different behaviour for energies above and below the chemical potential. For $E > \mu$, the exponential is very large, and the distribution function goes to zero. For $E < \mu$, the exponential is very small, and the distribution function tends to 1. This leads to a step-function behaviour, where we see distributions below μ of 1 and no occupation of energy levels above μ. This is again explained through the quantum statistics: fermions obey the Pauli exclusion principle and will fill the lowest levels, one fermion at a time. An illustration of this behaviour can be seen in Fig. 2.1.

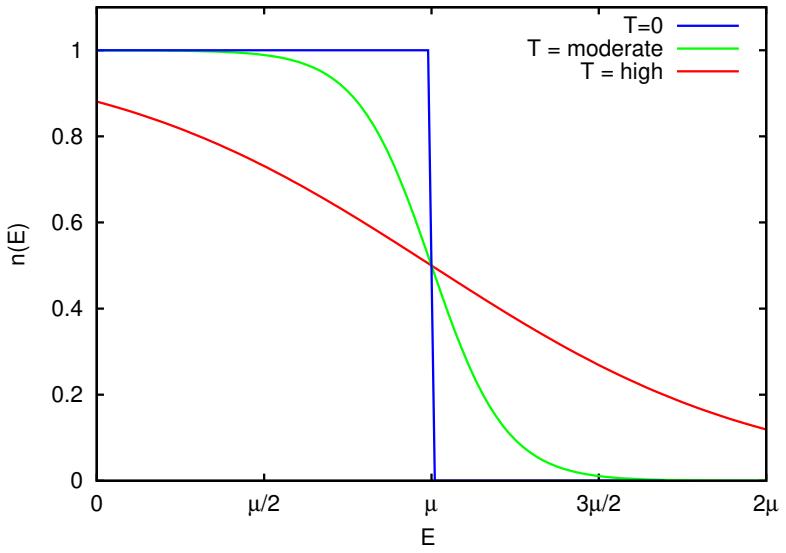

Figure 2.1: The behaviour of the fermion distribution function at different energies.

Chapter 3

Relativistic Quantum Mechanics

3.1 Relativistic wave equations

Core reading: Chapter 15.1 of Bransden & Joachain

Relativistic Quantum Mechanics is the union of two of the foundation stones of modern physics *Quantum Mechanics* (QM) and *Special Relativity* (SR). In this course we will study two of the most important relativistic wave equations: the Klein-Gordon and the Dirac equations. We will find that the union of QM and SR is far more interesting that just the construction of Lorentz invariant wave functions. Ultimately, our efforts will lead us to a deeper understanding of spin, and our struggles to interpret the equations we study will lead us to Dirac's prediction of antimatter.

3.1.1 Recap of Lorentz transformations

We will briefly discuss Lorentz transformations, partly as a recap and partly to fix our notation. The space-time vectors are given as a *contra-variant vector*

$$x^\mu = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix} = \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix} \tag{3.1}$$

or as a *co-variant vector*

$$x_\mu = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} ct \\ -x \\ -y \\ -z \end{pmatrix} . \tag{3.2}$$

The metric tensor is given by

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} , \tag{3.3}$$

and we can use the metric tensor to raise and lower indicies,

$$g_{\mu\nu} = g^{\mu\nu} , \tag{3.4}$$

$$x_\mu = g_{\mu\nu} x^\nu , \tag{3.5}$$

$$x^\mu = g^{\mu\nu} x_\nu , \tag{3.6}$$

$$g^\mu_\nu = g^{\mu\sigma} g_{\sigma\nu} = \delta^\mu_\nu . \tag{3.7}$$

Remember that greek indices will be taken to be spacetime indices, $\mu \in \{0, 1, 2, 3\}$, and that we will regularly use Einstein summation notation where repeated indices are summed over this range.

A *Lorentz transformation* is a transformation between two inertial frames: non-accelerating systems that move relatively to each other with a constant velocity. This is a linear transformation on the spacetime coordinates, and as such can be expressed by a matrix Λ, which in component notation reads

$$(x')^\mu = \Lambda^\mu{}_\nu x^\nu.$$

where x' is the 4-vector spacetime position in the new frame. Any 4-component quantity which transforms in the same way as x^μ is known as a 4-vector, *e.g* p^μ the energy-momenutm 4-vector. The inner product between two 4-vectors, that is the sum over all values of indices for the products of a contravariant and covariant object, is *invariant* under Lorentz transformations. If we introduce two arbitrary 4-vectors a^μ and b^μ, the inner product is given by

$$a^\mu b_\mu = g_{\mu\nu} a^\mu b^\nu$$

Ensuring that this is invariant under Lorentz transformations requires,

$$\begin{aligned} a^\mu b_\mu &= (a')^\mu (b')_\mu, \\ a^\alpha g_{\alpha\beta} b^\beta &= g_{\mu\nu} (a')^\mu (b')^\nu, \\ &= g_{\mu\nu} \Lambda^\mu{}_\alpha a^\alpha \Lambda^\nu{}_\beta b^\beta, \\ &= a^\alpha \left(\Lambda^{\mathrm{T}} g \Lambda\right)_{\alpha\beta} b^\beta. \end{aligned}$$

Note that we've used the freedom to redefine dummy variables in the Einstein summation notation in the manipulations above. We see that for the inner product to be an invariant, we require

$$g_{\alpha\beta} = \left(\Lambda^{\mathrm{T}} g \Lambda\right)_{\alpha\beta}.$$

This can be taken as the defining relation for the Lorentz transformation matrix Λ. A systematic study of the different types of Lorentz transformation is beyond the scope of this course; however, it should be a familiar fact that the set of all Lorentz transformations can be divided into a number of important sub-sets,

- Rotations *e.g.* a standard rotation through an angle ϕ about the z axis

$$R_{12} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\phi & \sin\phi & 0 \\ 0 & -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{3.8}$$

- Boosts *e.g.* a one-dimensional Lorentz transformation in the x direction

$$L_{01} = \begin{pmatrix} \cosh\eta & -\sinh\eta & 0 & 0 \\ -\sinh\eta & \cosh\eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{1-\beta^2}} & \frac{-\beta}{\sqrt{1-\beta^2}} & 0 & 0 \\ \frac{-\beta}{\sqrt{1-\beta^2}} & \frac{1}{\sqrt{1-\beta^2}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{3.9}$$

  with $\beta = v/c$ and $\tanh\eta = \beta$. The first form of this matrix uses the *rapdity* η to define the Lorentz boost. If this is unfamiliar to you, look it up in a textbook or online. It is identical to the description of a Lorentz transformation in terms of γ and β given in the second version of the matrix, but it highlights the analogy between boosts and rotations. See also the written solutions for Problem 9 where a few details of this notation are discussed.

- Spatial inversions (parity transformations, $\underline{x} \to -\underline{x}$)

- Time reversal (the mapping $t \to -t$)

These latter two form an imporant class of *discrete spacetime symmetries* which are very important for the study of fundamental particle interactions. In fact, the discovery that the weak force violates parity symmetry ($\underline{x} \to -\underline{x}$) led to a Nobel prize for T.-D. Lee and C.N. Yang in 1957, and plays a crucial role in the Standard Model of particle physics. We will restrict ourselves to study the *proper orthochronous Lorentz transformations*. Which further obey $\det\Lambda = +1$ and $\Lambda^0{}_0 \geq 1$. These constraints exclude the discrete spacetime symmetries, and this set of transformations consists only of three-dimensional rotations and boosts.

Note that the differential operator ∂_μ has the same transformation property as x_μ,

$$\partial_\mu = \frac{\partial}{\partial x^\mu} = \frac{\partial (x')^\nu}{\partial x^\mu}\frac{\partial}{\partial (x')^\nu} \tag{3.10}$$

$$= \Lambda^\nu{}_\mu \frac{\partial}{\partial (x')^\nu} = \Lambda^\nu{}_\mu \left(\partial'\right)_\nu \ . \tag{3.11}$$

Using this fact, we can verify a bit more formally that the *d'Alembertian*

operator is indeed Lorentz invariant,

$$\begin{aligned}
\Box = \partial_\mu \partial^\mu &= \partial_\mu g^{\mu\nu} \partial_\nu && (3.12)\\
&= \Lambda^\alpha{}_\mu g^{\mu\nu} \Lambda^\beta{}_\nu (\partial')_\alpha (\partial')_\beta && (3.13)\\
&= g^{\alpha\beta} (\partial')_\alpha (\partial')_\beta \ , && (3.14)\\
&= (\partial')_\alpha (\partial')^\alpha = \Box'. && (3.15)
\end{aligned}$$

3.1.2 The Klein-Gordon equation

So far, all quantum systems that we have studied have been described by the Schrödinger equation,

$$H\Psi = \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\underline{r}) \right] \Psi = i\hbar \frac{\partial \Psi}{\partial t}.$$

The form of this equation can be 'derived' using the *correspondence principle*. This says that to find a quantum mechanical analogue of a classical system, we should find a classical equation for the Hamiltonian and replace the classical observables with quantum operators. Specifically, we use the replacements

$$E \to i\hbar \frac{\partial}{\partial t}, \qquad \text{and} \qquad \underline{p} \to -i\hbar \underline{\nabla},$$

in the coordinate representation. The Schrödinger equation can be derived by applying this principle to the classical Hamiltonian operator:

$$\begin{aligned}
H &= T + V, \\
&= \frac{1}{2} m v^2 + V(\underline{r}), \\
&= \frac{p^2}{2m} + V(\underline{r}).
\end{aligned}$$

Which leads to the quantum mechanical operator as expected,

$$H = -\frac{\hbar^2}{2m} \nabla^2 + V(\underline{r}).$$

This process was explicitly non-relativistic. Our expression for the energy of a particle, expressed through its classical Hamiltonian, relied on the identification of the kinetic energy $T = \frac{1}{2} m v^2$. Consequently, the Schrödinger equation is only valid for non-relativistic situations. Another way to see that the Schrödinger equation is not a relativistic wave equation, is to note that time and space appear on a very different footing: time appears as a

first derivative (related to the energy) but space appears as a second derivative (related to the momentum squared). Without a sense of space-time symmetry, we cannot expect our theory to be relativistically invariant.

We could expect to derive an explicitly relativistic quantum mechanical operator by applying the correspondence principle to the relativistic energy-momentum relationship:

$$E^2 = \underline{p}^2 c^2 + m^2 c^4.$$

Under our correspondence, this produces the operator

$$-\hbar^2 \frac{\partial^2}{\partial t^2} = -\hbar^2 c^2 \underline{\nabla}^2 + m^2 c^4.$$

Acting this operator on a wavefunction $\Phi(\underline{x}, t)$ produces

$$\left[\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \underline{\nabla}^2 + \frac{m^2 c^2}{\hbar^2} \right] \Phi(\underline{x}, t) = 0.$$

This equation is known as the *Klein-Gordon equation.* To emphasise its space-time symmetry, we can write it in a covariant notation using 4-vectors. If you recall the derivative operator on space time can be expressed by

$$\partial_\mu := \frac{\partial}{\partial x^\mu},$$

we can write the Klein-Gordon equation as

$$\left(\partial_\mu \partial^\mu + \frac{m^2 c^2}{\hbar^2} \right) \Phi(\underline{x}, t) = 0. \tag{3.16}$$

The contraction of the two partial derivatives is the *d'Alembertian* operator, which is often given its own symbol,

$$\Box := \partial_\mu \partial^\mu = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \underline{\nabla}^2.$$

If we use natural units ($\hbar = c = 1$) and the definition of the d'Alembertian operator, the Klein-Gordon equation takes the compact form,

$$\left(\partial_\mu \partial^\mu + m^2 \right) \Phi(\underline{x}, t) = 0. \tag{3.17}$$

We note two things about the Klein-Gordon operator. First, it has a sense of symmetry that the Schrödinger equation did not have: the time derivative now appears as a second derivative, as do the spatial derivatives. Moreover, the equation is truly Lorentz invariant as it only depends on the spacetime indices through the d'Alembertian which has no free indices. This is reassuring, as we chose to start with a Lorentz invariant energy-momentum relation exactly to enforce this property.

3.1.3 Charged particle in an EM field

We will now consider how the Klein-Gordon equation changes when we look at the low-energy, non-relativistic limit. However, it is more interesting to consider this problem for a particle coupled to an electromagnetic (EM) field. We introduce the coupling of the particle to the EM field by using *minimal coupling*. This involves describing the field by its scalar and vector potentials,

$$A^\mu = \begin{pmatrix} \frac{1}{c}\phi \\ \underline{A} \end{pmatrix},$$

for a scalar potential ϕ and a 3-vector potential $\underline{A}$. I have used common notation to expresses a 4-vector A^μ as a scalar part and a 3-vector part, take a moment to make sure you are happy with notation such as this. The minimal coupling prescription replaces energies and momenta with the combinations

$$E \to E - q\phi \qquad \text{and} \qquad \underline{p} \to \underline{p} - q\underline{A}.$$

This leads us to the minimally coupled Klein-Gordon equation,

$$(E - q\phi)^2 \, \Phi(\underline{x}, t) = \left[m^2 c^4 + \left(\underline{p} - q\underline{A}\right)^2 c^2 \right] \Phi(\underline{x}, t),$$

which in the coordinate representation can be given as

$$\left(i\hbar \frac{\partial}{\partial t} - q\phi \right)^2 \Phi(\underline{x}, t) = \left[m^2 c^4 + (i\hbar \underline{\nabla} + q\underline{A})^2 c^2 \right] \Phi(\underline{x}, t). \tag{3.18}$$

We will now show that in the low-energy limit, this faithfully reproduces the Schrödinger equation coupled to an EM field. However, the correspondence is not exact: the relativistic energy E contains a contribution from the rest mass of the particle. At low-energies, the energy of the particle is dominated by this large contribution and the classical kinetic energy arises as the first correction

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - \frac{v^2}{c^2}}} = mc^2 + \frac{1}{2}mv^2 + \mathcal{O}(\beta^4).$$

Therefore we expect the quantity $\widetilde{E} = E - mc^2$ to correspond to the non-relativistic energy, not E. We redefine our wavefunction to remove this dependence on the rest-mass energy,

$$\Phi(\underline{x}, t) =: e^{-i\frac{mc^2}{\hbar}t} \widetilde{\Phi}(\underline{x}, t).$$

Plugging this into Eq. (3.18), we find

$$\left(i\hbar\frac{\partial}{\partial t} - q\phi\right)^2 e^{-i\frac{mc^2}{\hbar}t}\widetilde{\Phi}(\underline{x},t) = \left(i\hbar\underline{\nabla} + q\underline{A}\right)^2 c^2 e^{-i\frac{mc^2}{\hbar}t}\widetilde{\Phi}(\underline{x},t). \tag{3.19}$$

We deal with the left-hand side first, expanding the bracket and using the chain rule,

$$\begin{aligned}
\text{LHS} &\equiv \left[-\hbar^2\frac{\partial^2}{\partial t^2} + q^2\phi^2 - iq\hbar\frac{\partial\phi}{\partial t} - 2iq\hbar\phi\frac{\partial}{\partial t}\right] e^{-i\frac{mc^2}{\hbar}t}\widetilde{\Phi}(\underline{x},t), \\
&\equiv \left(-\hbar^2\frac{\partial}{\partial t} - 2iq\hbar\phi\right)\left[-i\frac{mc^2}{\hbar}\widetilde{\Phi}(\underline{x},t) + \frac{\partial\widetilde{\Phi}}{\partial t}\right] e^{-i\frac{mc^2}{\hbar}t} + \left[q^2\phi^2 - iq\hbar\frac{\partial\phi}{\partial t}\right] e^{-i\frac{mc^2}{\hbar}t}\widetilde{\Phi}(\underline{x},t), \\
&\equiv \left[-\hbar^2\frac{\partial^2\widetilde{\Phi}}{\partial t^2} + 2i\hbar(mc^2 - q\phi)\frac{\partial\widetilde{\Phi}}{\partial t} + \left(q^2\phi^2 - iq\hbar\frac{\partial\phi}{\partial t} - 2q\phi mc^2\right)\widetilde{\Phi}(\underline{x},t)\right] e^{-i\frac{mc^2}{\hbar}t}.
\end{aligned}$$

We now make some assumptions about the relative size of these terms,

$$\begin{aligned}
|q\phi| &\ll mc^2, \\
\left|\hbar\frac{\partial\phi}{\partial t}\right| &\ll mc^2\phi, \\
\left|\hbar\frac{\partial^2\widetilde{\Phi}}{\partial t^2}\right| &\ll \left|mc^2\frac{\partial\widetilde{\Phi}}{\partial t}\right|,
\end{aligned}$$

which correspond to the assumption that the wavefunction is slowly evolving and the EM field not sufficiently strong to invalidate the non-relativistic approximation. With these assumed hierarchies, we arrive at

$$\text{LHS} = \left[2i\hbar mc^2\frac{\partial\widetilde{\Phi}}{\partial t} - 2q\phi mc^2\widetilde{\Phi}(\underline{x},t)\right] e^{-i\frac{mc^2}{\hbar}t}.$$

Going back to Eq. (3.19) and cancelling some common factors, we arrive at

$$\left(i\hbar\frac{\partial}{\partial t} - q\phi\right)\widetilde{\Phi}(\underline{x},t) = \frac{\left(i\hbar\underline{\nabla} + q\underline{A}\right)^2}{2m}\widetilde{\Phi}(\underline{x},t).$$

This is precisely the Schrödinger equation for a particle minimally coupled to an EM field given by ϕ and $\underline{A}$.

3.1.4 Solutions to the Klein-Gordon equation

We can derive solutions to the Klein-Gordon equation without too much trouble. This will be a problem you study in Problem 1. See the written solutions for more discussion.

3.1.5 Interpretation of solutions

By using the correspondence principle, we have constructed a relativistic wave equation known as the Klein-Gordon equation. This seems to behave as expected: we have seen that it can be written in a Lorentz invariant way, and that its low-energy approximation is the Schrödinger equation from non-relativistic quantum mechanics. However as we will see in this section, its interpretation is not without difficulty and these challenges ultimately led to it being rejected as a suitable foundation for a relativistic theory of single particle quantum mechanics.

Probability current

Just as for the Schrödinger equation we can construct a conserved quantity which we hope to interpret as the probability current. You will work through the details in Problem 2, but it can be shown that we can derive a *continuity equation*

$$\underline{\nabla} \cdot \underline{j} = \frac{\partial \rho}{\partial t},$$

for two quantities ρ and $\underline{j}$, which are defined by

$$\rho = \Phi^* \frac{\partial \Phi}{\partial t} - \Phi \frac{\partial \Phi^*}{\partial t} \qquad \text{and} \qquad \underline{j} = \Phi \underline{\nabla} \Phi^* - \Phi^* \underline{\nabla} \Phi.$$

It can also be shown (see Problem 2), that these quantities reduce appropriately to the probability density and probability current for the Schrödinger equation in the non-relativistic limit. This suggests that the conservation equation that we have found is a relativistic analogue of the conservation of the probability current.

However, this is not such a straightforward identification. The 'probability density' ρ that we defined above is not necessarily a positive quantity. This would allow the system to occupy states with negative probability. This is conceptually very confusing, and the appearance of negative probabilities is a major failing of our interpretation of this equation. This problem led many to doubt the consistency of the Klein-Gordon particle as a relativistic quantum mechanical theory.

Free solutions and negative energies

We can find solutions to the free (without a potential) Klein-Gordon equation with a plane wave Ansatz,

$$\Phi(\underline{x}, t) = A e^{i(\underline{k} \cdot \underline{r} - \omega t)}. \tag{3.20}$$

where $\underline{k}$ denotes a wavevector, ω a frequency and A is an arbitrary complex constant. This provides a solution only if ω and $\underline{k}$ satisfy,

$$\omega^2 - \underline{k}^2 c^2 = \frac{m^2 c^4}{\hbar^2},$$

which has the solutions

$$\omega = \pm\sqrt{\underline{k}^2 c^2 + \frac{m^2 c^4}{\hbar^2}}. \tag{3.21}$$

The energy of this particle is given by $E = \hbar\omega$, and the negative values of ω would imply negative energy solutions. This is another major interpretational hurdle for a theory of matter, where negative energies are associated with unstable systems[1].

These interpretational issues led to the Klein-Gordon equation being largely discarded as a viable quantum theory. However, nowadays it plays an important role in Quantum Field Theory where the problems of negative energies and probability densities can be resolved. Although not proven in this course, the Klein-Gordon equation actually describes spin-0 particles, such as the pion or Higgs boson.

[1] We will discuss the conceptual issues related to negative particles in more detail when we discuss the interpretation of the Dirac equation in Lecture 17

3.2 The Dirac equation

Core reading: Chapter 15.2 of Bransden & Joachain

The Klein-Gordon equation was found by applying the correspondence principle to the relativistic energy-momentum relation,

$$E^2 - \underline{p}^2 c^2 = m^2 c^4,$$

which leads to

$$\left(\partial_\mu \partial^\mu + \frac{m^2 c^2}{\hbar^2}\right) \Phi(\underline{x}, t) = 0.$$

This equation applies to spin-0 particles (which is not something I've proven for you), but it leads to some apparent inconsistencies in its interpretation.

Paul Dirac (1902–1984), one of the most important physicists of the last 100 years, thought that some of these inconsistencies might lie in the fact that it was a *second-order* differential equation in time. This second-order dependence on time would always imply that the energy would have multiple solutions related to the multivalued nature of the square root, just as with the energy-momentum relationship

$$E = \pm\sqrt{\underline{p}^2 c^2 + m^2 c^4}.$$

He was therefore motivated to construct a Lorentz invariant *first-order* wave equation. Dirac may well have objected to the characterisation of his work that I've just presented. He was known to eschew such physically based motivations, claiming that

> *I think it's a peculiarity of myself that I like to play about with equations, just looking for beautiful mathematical relations which maybe don't have any physical meaning at all. Sometimes they do.*
>
> – P. Dirac in an interview with Thomas Kuhn (1963).

We will see that Dirac did indeed find beautiful mathematical structures in his work on relativistic equations, and despite his feigned indifference to experimental confirmation, we'll see that his theories have led to some of the most precise agreements between experiment and theory in the entirety of the history of science.

3.2.1 Derivation

We follow a historical route to derive the Dirac equation. We start with an Ansatz for the Hamiltonian proposed by Dirac,

$$\begin{aligned} i\hbar\frac{\partial\Psi}{\partial t} &= H_{\mathrm{D}}\Psi(\underline{x},t), \\ &= -i\hbar c\sum_{k=1}^{3}\alpha^k\frac{\partial\Psi}{\partial x^k} + \beta mc^2\Psi(\underline{x},t), \end{aligned}$$

where the quantities α^k and β are unknowns that Dirac hoped to find. This equation, known as the Dirac equation, might look like a complicated first guess, but behind the $\hbar$'s and c's it is simply the most general linear combination of the three spatial derivatives, which correspond to momentum operators, and an additional constant term proportional to the rest mass of the particle,

$$H_{\mathrm{D}} = \underline{\alpha}\cdot\underline{p}c + \beta mc^2.$$

for $p_k = -i\hbar\nabla_k$.

We know that any relativistic particle must satisfy the energy-momentum relationship, and this places a constraint on H_{D}. We require that for all solutions to Dirac's first-order Ansatz, that

$$\begin{aligned} E^2\Psi(\underline{x},t) &= \left(p^2c^2 + m^2c^4\right)\Psi(\underline{x},t), \\ \implies H_{\mathrm{D}}^2\Psi(\underline{x},t) &= \left(-\hbar^2c^2\underline{\nabla}^2 + m^2c^4\right)\Psi(\underline{x},t). \end{aligned} \tag{3.22}$$

We can evaluate the left-hand side of this equation using Dirac's Ansatz for H_{D},

$$\begin{aligned} H_{\mathrm{D}}^2 &= \left(-i\hbar c\alpha^j\partial_j + \beta mc^2\right)\left(-i\hbar c\alpha^k\partial_k + \beta mc^2\right), \\ &= -\hbar^2c^2\alpha^j\alpha^k\partial_j\partial_k - i\hbar c^3m\alpha^j\beta\partial_j - i\hbar c^3m\beta\alpha^k\partial_k + \beta^2m^2c^4, \\ &= -\frac{\hbar^2c^2}{2}\left\{\alpha^j,\alpha^k\right\}\partial_j\partial_k - i\hbar c^3m\left\{\alpha^i,\beta\right\}\partial_i + \frac{m^2c^4}{2}\left\{\beta,\beta\right\}. \end{aligned}$$

Note that we have done something slightly unusual in this last line: we have not assumed that the quantities α^k and β commute. Normal numbers, be they real or complex, clearly commute. Dirac took the surprising step of allowing these quantities to be something new. We have manipulated these expressions without assuming that they commute and expressed them in terms of the *anti-commutator*

$$\{A,B\} = AB + BA.$$

We see that our expression for H_{D}^2 will only satisfy the energy-momentum relationship in Eq. (3.22) if the α^k and β quantities obey

$$\begin{aligned}\left\{\alpha^j, \alpha^k\right\} &= 2\delta^{jk}, \\ \left\{\alpha^i, \beta\right\} &= 0, \\ \{\beta, \beta\} &= 2.\end{aligned}$$

These algebraic relations are fundamental to the study of Dirac equation and are known as the *Dirac algebra*.

To solve these algebraic relations we need to consider non-commuting objects. A familiar example of not-necessarily-commuting quantities is a set of matrices, and matrices which satisfy the above relations are known as *Dirac matrices*. There are many different sets of Dirac matrices[2], but we will use the following

$$\alpha^k = \begin{pmatrix} 0 & \sigma^k \\ \sigma^k & 0 \end{pmatrix} \qquad \text{and} \qquad \beta = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}.$$

which are 4-dimensional matrices written in 2×2 block notation. Each entry should be interpreted as a 2×2 matrix: σ^k denotes the k-th Pauli matrix, I_2 is the 2×2 identity matrix and 0 is a 2×2 block of zeros. To give some explicit examples,

$$\alpha^1 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} \qquad \text{and} \qquad \beta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

This is a common notation, and I will use it often, please take a moment to make sure you understand how to go from one notation to the other. Also, see the written solutions for the problem sets and workshop problems where you can see the notation in concrete applications.

Some properties of the Dirac Hamiltonian

- The matrices I have given above are 4-dimensional, and it is a remarkable fact that this is the smallest dimension for which we can find matrices which satisfy the Dirac algebra. You will prove this fact yourself in Problem 4. See the solutions for more discussion.

[2] We'll discuss later how we can change the form of the Dirac matrices while still satisfying the algebraic relations.

- All Hamiltonians and Hermitian[3]. Imposing Hermiticity constrains the matrices α^k and β to also be Hermitian,

$$\left(\alpha^k\right)^\dagger = \alpha^k \qquad \text{and} \qquad \beta^\dagger = \beta.$$

- As mentioned before, the representation of the Dirac matrices is not unique. We can see this quite easily by showing that for any set of Dirac matrices, α^k and β, we can define another set of matrices

$$\widetilde{\alpha}^k = M\alpha^k M^{-1} \qquad \text{and} \qquad \widetilde{\beta} = M\beta M^{-1}.$$

  which will satisfy the Dirac algebra,

$$\{\widetilde{\alpha}^j, \widetilde{\alpha}^k\} = 2\delta^{jk}, \qquad \{\widetilde{\alpha}^j, \widetilde{\beta}\} = 0 \quad \text{and} \quad \{\widetilde{\beta}, \widetilde{\beta}\} = 2.$$

  The proof is left as an exercise. (Although, there are close parallels to part (b) of Problem 5. For the new matrices to be Hermitian, we require

$$\begin{aligned}\widetilde{\beta}^\dagger &= \left(M\beta M^{-1}\right)^\dagger, \\ &= \left(M^{-1}\right)^\dagger \beta^\dagger M^\dagger, \\ &= \left(M^{-1}\right)^\dagger \beta M^\dagger.\end{aligned}$$

  For this to be satisfied we require $M^\dagger = M^{-1}$, or in other words, that M is unitary. Therefore, any unitary matrix can be used to produce a new set of Dirac matrices from an old set.

- Note that as the objects α^k and β are now 4-dimensional matrices, the wavefunction must also be an object with multiple components. This is known as a *Dirac spinor* and it has 4 complex components whose evolution is coupled through the Dirac equation,

$$\Psi = \begin{pmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \\ \Psi_4 \end{pmatrix}.$$

[3] Like all dogmatic statements, in some unconventional (and off-syllabus) settings this may not be true.

3.2.2 Continuity equation

The negative 'probability density' of the Klein-Gordon equation was an important motivation for the construction of the Dirac equation. We'll now see to what extent this problem was circumvented.

We can construct a conservation equation in the same manner as for the Schrödinger equation, by first constructing a probability density and taking its time derivative. We define

$$\begin{aligned}
\rho &= \Psi^\dagger \Psi, \\
&= (\Psi_1^*, \Psi_2^*, \Psi_3^*, \Psi_4^*) \begin{pmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \\ \Psi_4 \end{pmatrix}, \\
&= |\Psi_1|^2 + |\Psi_2|^2 + |\Psi_3|^2 + |\Psi_4|^2, \\
&\geq 0.
\end{aligned}$$

Now we take the time derivative,

$$\frac{\partial \rho}{\partial t} = \Psi^\dagger \frac{\partial \Psi}{\partial t} + \frac{\partial \Psi^\dagger}{\partial t} \Psi. \tag{3.23}$$

These two terms can both be simplified using the Dirac equation, first we derive an identity for the derivative of Ψ,

$$\begin{aligned}
\Psi^\dagger \frac{\partial \Psi}{\partial t} &= \Psi^\dagger \frac{1}{i\hbar} H_{\mathrm{D}} \Psi, \\
&= \Psi^\dagger \left(-c\alpha^k \nabla_k + \frac{mc^2}{i\hbar} \beta \right) \Psi, \\
&= -c\Psi^\dagger \underline{\alpha} \cdot \underline{\nabla} \Psi + \frac{mc^2}{i\hbar} \Psi^\dagger \beta \Psi.
\end{aligned} \tag{3.24}$$

Taking the complex conjugate of this expression we find another relation,

$$\frac{\partial \Psi^\dagger}{\partial t} \Psi = -c \left(\underline{\nabla} \Psi^\dagger \right) \cdot \underline{\alpha} \Psi - \frac{mc^2}{i\hbar} \Psi^\dagger \beta \Psi. \tag{3.25}$$

We use equations Eqs. (3.24) and (3.25) in Eq. (3.23) to find,

$$\begin{aligned}
\frac{\partial \rho}{\partial t} &= -c \left(\underline{\nabla} \Psi^\dagger \right) \cdot \underline{\alpha} \Psi - c\Psi^\dagger \underline{\alpha} \cdot \underline{\nabla} \Psi, \\
\frac{\partial \rho}{\partial t} + \underline{\nabla} \cdot \left(\Psi^\dagger c \underline{\alpha} \Psi \right) &= 0, \\
\frac{\partial \rho}{\partial t} + \underline{\nabla} \cdot \underline{j} &= 0,
\end{aligned}$$

where in the final line we have made the identification,

$$\underline{j} = \Psi^\dagger c\underline{\alpha}\Psi.$$

We have therefore found a continuity equation, with a positive definite probability density ρ. Dirac's brave derivation of a multi-component wave equation has appeared to solve one of the difficulties associated with the Klein-Gordon equation.

3.2.3 Covariant form of the Dirac equation

We have followed the historical derivation of the Dirac equation and introduced factors α^k for the momenta and a matrix β associated with the rest mass of the particle. However, this notation hides some of the symmetry between space and time, and ultimately complicates the analysis of the Dirac equation under Lorentz transformations. In this section we will introduce an alternative notation, known as the *covariant form* of the Dirac equation, which increases the aesthetic symmetry between space and time.

We start by noting that

$$\{\beta, \beta\} = 2 \qquad \Longrightarrow \qquad \beta^2 = I_4,$$

where I_4 is the 4×4 identity matrix. This allows us to rewrite the Dirac equation in the following way

$$\begin{aligned} i\hbar\frac{\partial\Psi}{\partial t} &= -i\hbar c\alpha^k\frac{\partial\Psi}{\partial x^k} + \beta mc^2\Psi(\underline{x}, t), \\ \Longrightarrow \qquad -i\beta\frac{1}{c}\frac{\partial\Psi}{\partial t} &= i\beta\alpha^k\frac{\partial\Psi}{\partial x^k} - \frac{mc}{\hbar}\Psi(\underline{x}, t). \end{aligned}$$

Using the notation of the covariant derivative

$$\partial_\mu = \left(\frac{1}{c}\frac{\partial}{\partial t}, \frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3}\right),$$

we can write this in the form

$$\left(-i\gamma^\mu\partial_\mu + \frac{mc}{\hbar}\right)\Psi(\underline{x}, t) = 0.$$

where we have introduced new 4×4 matrices,

$$\gamma^0 = \beta \qquad \text{and} \qquad \gamma^k = \beta\alpha^k.$$

These matrices are known as *gamma matrices*. They are an alternative way of representing the Dirac algebra.

It is left as an exercise to prove the following important facts:

- Only one of the gamma matrices is Hermitian. The others are anti-Hermitian,

$$\left(\gamma^0\right)^\dagger = \gamma^0 \qquad \text{and} \qquad \left(\gamma^k\right)^\dagger = -\gamma^k.$$

- The Dirac algebra can be rewritten in terms of the gamma matrices,

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu},$$

where $g^{\mu\nu}$ is the Minkowski metric,

$$g^{\mu\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}.$$

Note that this expression of the Dirac algebra has two different 4-dimensional sets of indices in it. Do not get them confused! Each γ^μ, where μ is some fixed integer denoting a spacetime direction, is a 4×4 matrix in spinor space. We could write the indices of these matrices explicitly,

$$\left(\gamma^\mu\right)_{ab},$$

where $a, b \in \{1, 2, 3, 4\}$ denote spinor indices but $\mu \in \{0, 1, 2, 3\}$ denotes a spacetime direction. We can write the Dirac algebra relation with these indices for clarity,

$$\begin{aligned} \{\gamma^\mu, \gamma^\nu\}_{ab} &= 2g^{\mu\nu}\delta_{ab}, \\ \left(\gamma^\mu\gamma^\nu\right)_{ab} + \left(\gamma^\nu\gamma^\mu\right)_{ab} &= 2g^{\mu\nu}\delta_{ab}, \\ \sum_{c=1}^{4} \left(\gamma^\mu\right)_{ac} \left(\gamma^\nu\right)_{cb} + \sum_{c=1}^{4} \left(\gamma^\nu\right)_{ac} \left(\gamma^\mu\right)_{cb} &= 2g^{\mu\nu}\delta_{ab}. \end{aligned}$$

In the first line above we have added a δ_{ab} matrix in spinor space to make sure the right-hand side of the Dirac algebra relation is a spinor matrix. These factors are often omitted, but you must remember that they are there. This could be written as

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I, \tag{3.26}$$

where I is the identity matrix in spinor space, $I_{ab} = \delta_{ab}$.

- Just like the α^k and β matrices, we can always form new gamma matrices which satisfy the Dirac algebra by multiplying them by a unitary matrix. See Problem 5, part (b) for details. This means that there is no unique form of the gamma matrices. One important form of the gamma matrices is known as the *chiral representation*,

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} \qquad \text{and} \qquad \gamma^k = \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix} \tag{3.27}$$

  The above expression again uses the gamma matrices in 2×2 block form: σ^k is the k-th Pauli matrix and I_2 is the 2×2 identity matrix.

- With one more notational convention we can express the Dirac equation in an even more aesthetically pleasing form

$$\left(i\partial\!\!\!/ - \frac{mc}{\hbar}\right) \Psi(\underline{x}, t) = 0,$$

  where $\partial\!\!\!/ =: \gamma^\mu \partial_\mu$. This is known as *Feynman slash notation* after Richard Feynman (1918–1988) one of the pioneers of Quantum Field Theory. We can use this notation on any four vector, let's denote an arbitrary 4-vector by a_μ, and the slashed version of this quantity $a\!\!\!/$ is just the 4-vector contracted with the gamma matrices,

$$a\!\!\!/ = \gamma^\mu a_\mu.$$

- If we use natural units ($\hbar = c = 1$) and the Feynman slash, the Dirac equation takes the compact form

$$\left(i\partial\!\!\!/ - m\right) \Psi = 0. \tag{3.28}$$

  This is (rightly) regarded as one of the most beautiful equations in physics. It is inscribed on Dirac's memorial stone in Westminster Abbey.

3.3 Lorentz covariance of the Dirac equation

Core reading: Chapter 15.3 of Bransden & Joachain

We have seen that the Dirac equation can be written in a *covariant form*,

$$\left(i\partial\!\!\!/ - m\right)\Psi(\underline{x}, t) = 0,$$

where in this chapter we will use natural units $\hbar = c = 1$, and we have used the Feynman slash notation on the 4-derivative,

$$\partial\!\!\!/ := \gamma^\mu \partial_\mu.$$

This form of the Dirac equation certainly looks like it might be Lorentz covariant, it treats space and time in a manifestly similar way; however, we should note that despite our notation, the set of gamma matrices γ^μ is an unusual 4-component object. We gave an explicit form of the gamma matrices in the chiral representation,

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix} \qquad \text{and} \qquad \gamma^k = \begin{pmatrix} 0 & \sigma^k \\ -\sigma^k & 0 \end{pmatrix},$$

where, as before, these 4×4 matrices are given in 2×2 block form: σ^k is the k-th Pauli matrix and I_2 is the 2×2 identity matrix. However, despite our notation, we would not expect this object to be a 4-vector. In fact, the form of the gamma matrices given above is *not a frame dependent quantity*: in all inertial frames the representation remains unchanged. In this section, we'll study how the system changes under Lorentz transformation and how Dirac spinors must transform for the Dirac equation to remain true in all frames.

3.3.1 Lorentz covariance of the Dirac equation

Now we want to see how the Dirac equation transforms when performing a Lorentz transformation. The only additional complication is that we must also allow for the Dirac spinor Ψ to transform as we change frame. In general, this will also be a linear transformation, and can be represented by a matrix, but now this matrix will act on the spinor indices of Ψ. We denote this transformation as

$$\begin{aligned} x' &= \Lambda x \ , \\ \Psi'(x') &= S(\Lambda)\Psi(x) \ , \end{aligned}$$

where $S(\Lambda)$ is a 4×4 matrix in spinor space which will depend on the specific Lorentz transformation in question.

We denote the initial frame with unprimed quantites and the boosted frame with primed quantities,

System I :	Coordinate x^μ	spinor Ψ
System I' :	Coordinate $(x')^\mu$	spinor Ψ'

(3.29)

Starting from the Dirac equation in System I

$$\left(-i\gamma^\mu\partial_\mu + \frac{mc}{\hbar}\right)\Psi(x) = 0 \,, \tag{3.30}$$

we transform ∂_μ and $\Psi(x)$ into the system I'

$$\partial_\mu = \Lambda^\nu{}_\mu \partial'_\nu \,, \tag{3.31}$$

$$\Psi(x) = S(\Lambda)^{-1}\Psi'(x') \,, \tag{3.32}$$

to get

$$\left(-i\gamma^\mu\Lambda^\nu{}_\mu\partial'_\nu + \frac{mc}{\hbar}\right)S(\Lambda)^{-1}\Psi'(x') = 0 \tag{3.33}$$

$$\left(-iS(\Lambda)\gamma^\mu\Lambda^\nu{}_\mu S(\Lambda)^{-1}\partial'_\nu + \frac{mc}{\hbar}\right)\Psi'(x') = 0 \,. \tag{3.34}$$

In the second line we have multiplied everything with $S(\Lambda)$ from the right-hand side. The Dirac equation would have the same form in both frames (it would be *covariant*) if

$$S(\Lambda)\gamma^\mu\Lambda^\nu{}_\mu S(\Lambda)^{-1} = \gamma^\nu \tag{3.35}$$

$$\Leftrightarrow \gamma^\mu\Lambda^\nu{}_\mu = S(\Lambda)^{-1}\gamma^\nu S(\Lambda) \,. \tag{3.36}$$

The relation in Eq. (3.36) can be seen as an analogous statement to the Lorentz condition,

$$g = \Lambda^{\mathrm{T}} g \Lambda,$$

and in the same way that we used the Lorentz condition to find the form of Λ, we will actually use Eq. (3.36) to define the matrix $S(\Lambda)$. That is to say, for a given Lorentz transformation we can find the matrix $S(\Lambda)$ by ensuring that this relation holds. An example of this calculation will be given in Problem 6. See the written solutions for more discussion. At this point, we will just give a few examples of the form of $S(\Lambda)$ for the Lorentz transformations presented in Eqs. (3.8) and (3.9):

- A spatial rotation around the z-axis:

$$
\begin{aligned}
R_{12} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\phi & \sin\phi & 0 \\ 0 & -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
\Rightarrow S(R_{12}) &= \begin{pmatrix} \cos\frac{\phi}{2} + i\sin\frac{\phi}{2} & 0 & 0 & 0 \\ 0 & \cos\frac{\phi}{2} - i\sin\frac{\phi}{2} & 0 & 0 \\ 0 & 0 & \cos\frac{\phi}{2} + i\sin\frac{\phi}{2} & 0 \\ 0 & 0 & 0 & \cos\frac{\phi}{2} - i\sin\frac{\phi}{2} \end{pmatrix}.
\end{aligned}
\tag{3.37}
$$

  Note that the matrix R_{12} has a periodicity of 2π, while $S_{R_{12}}$ has a periodicity of 4π. The spinors pick up a factor of (-1) after a rotation though $360°$.

- A Lorentz-boost along the x-axis

$$
\begin{aligned}
L_{01} &= \begin{pmatrix} \cosh\eta & -\sinh\eta & 0 & 0 \\ -\sinh\eta & \cosh\eta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
\Rightarrow S(L_{01}) &= \begin{pmatrix} \cosh\frac{\eta}{2} & 0 & \sinh\frac{\eta}{2} \\ 0 & \cosh\frac{\eta}{2} & \sinh\frac{\eta}{2} & 0 \\ 0 & \sinh\frac{\eta}{2} & \cosh\frac{\eta}{2} & 0 \\ \sinh\frac{\eta}{2} & 0 & 0 & \cosh\frac{\eta}{2} \end{pmatrix}.
\end{aligned}
\tag{3.38}
$$

3.3.2 Bilinear forms

We know that we can find quantities which are invariant under Lorentz transformations. For example, the contraction of two 4-vectors is the same in all frames,

$$a^\mu b_\mu = (a')^\mu (b')_\mu.$$

It would be interesting to look for Lorentz invariants composed of Dirac spinors. In fact, the construction of invariants of this kind will be very important when trying to represent physical quantities out of spinors.

We might think that a good object to start with is the product of the Dirac spinor and its Hermitian conjugate, which is analogous to the preseved length of a complex vector under complex rotations,

$$\Psi^\dagger \Psi.$$

This looks like a reasonable guess and we can quickly check that from the explicit form given above for a spinor transformation corresponding to a spatial rotation about the z-axis that

$$[S(R_{12})]^\dagger \, S(R_{12}) = 1,$$

and therefore $\Psi^\dagger \Psi$ is invariant under this transformation,

$$\Psi^\dagger \Psi = (\Psi')^\dagger \, [S(R_{12})]^\dagger \, S(R_{12}) \Psi' = (\Psi')^\dagger \Psi'.$$

Unfortunately, trying the same procedure with the Lorentz boost L_{01} shows that this combination of spinors is not invariant under all Lorentz tranformations. The problem is that the boost matrices are not Hermitian,

$$[S(L_{01})]^\dagger \, S(L_{01}) \neq 1.$$

With a bit of trial an error, we instead turn our attention to the *adjoint spinor*, which is defined as

$$\overline{\Psi} = \Psi^\dagger \gamma^0 \ . \tag{3.39}$$

It transforms as

$$\Psi' = S\Psi \quad \Rightarrow \quad \Psi'^\dagger = \Psi^\dagger S^\dagger \tag{3.40}$$

$$\Rightarrow \quad \overline{\Psi}' = \Psi^\dagger S^\dagger \gamma^0 \ . \tag{3.41}$$

One can show that[4]

$$S^\dagger \gamma^0 = b \gamma^0 S^{-1} \ , \tag{3.42}$$

with

$$b = \begin{cases} +1 & \text{for} \quad \Lambda^{00} \geq +1 \\ -1 & \text{for} \quad \Lambda^{00} \leq -1 \end{cases} \ . \tag{3.43}$$

Therefore, if we restrict ourselves to proper orthochronous Lorentz transformations, we get the transformation of the adjoint spinor to be

$$\overline{\Psi}' \quad = \quad \overline{\Psi} S^{-1} \ , \tag{3.44}$$

This means that we can form a Lorentz invariant combination from the spinor Ψ,

$$\overline{\Psi}\Psi = \overline{\Psi'}\Psi'.$$

This object transforms as a Lorentz scalar.

[4] see *e.g.* Chapter 6.2.3 of *Advanced Quantum Mechanics* by F. Schwabl.

We can generalise this analysis to find other combinations of pairs of spinors which have simple behaviours under Lorentz transformations. These objects are often called *bilinear forms*[5]. Knowledge of the more complicated bilinears is not part of the current course, and the proof of the following is left to the interested reader. Under the Lorentz transformation

$$\Psi' = S(\Lambda)\Psi,$$

the following relations hold,

$$\begin{array}{ccccccc} \overline{\Psi}' & & \Psi' & = & & \overline{\Psi} & & \Psi\ , \\ \overline{\Psi}' & \gamma^\mu & \Psi' & = & \Lambda^\mu{}_\alpha & \bar{\Psi} & \gamma^\alpha & \Psi\ , \\ \overline{\Psi}' & \sigma^{\mu\nu} & \Psi' & = & \Lambda^\mu{}_\alpha \Lambda^\nu{}_\beta & \overline{\Psi} & \sigma^{\alpha\beta} & \Psi\ , \end{array} \tag{3.45}$$

where we have defined

$$\sigma^{\mu\nu} := \frac{i}{2}\left[\gamma^\mu, \gamma^\nu\right]\ . \tag{3.46}$$

Note that the first object is the Lorentz scalar that we discussed before, the second behaves like a Lorentz 4-vector, and the final one is a Lorentz two-index tensor. Given the simple behaviours of these combinations of spinors, we can find Lorentz invariants by contracting the spactime indices with other Lorentz vectors. For example, the vector bilinear above can be contracted with the 4-vector potential of an electromagnetic field,

$$A_\mu \overline{\Psi}\gamma^\mu\Psi.$$

This object is invariant under Lorentz transformations and describes the interaction between the electron and photon in Quantum Electrodynamics. Another Lorentz invariant could be written,

$$\left(\overline{\Psi_1}\gamma^\mu\Psi_2\right)\left(\overline{\Psi_3}\gamma_\mu\Psi_4\right),$$

using four different spinors. Terms like these were the crucial ingredient in the first mathematical theory of the Weak interaction, and can explain for example the β-decay of nuclei or the decay of the muon.

You should understand the Lorentz transformation properties of spinors, the adjoint spinor and the scalar invariant. However, understanding how or why these Lorentz invariant objects correspond to the particle physics processes I have just mentioned is well and truly off-syllabus.

[5]This is a bad choice of name as these objects are not bilinear in Ψ (they are sesquilinear, like the Hilbert space inner product).

3.4 Solutions of the Dirac equation

Core reading: Chapter 15.4 to 15.6 of Bransden & Joachain

We motivated the Dirac equation by certain conceptual problems associated with a more obviously relativistically invariant wave equation, the Klein-Gordon equation. We have seen that the Dirac equation does indeed provide a relativistically covariant equation which allows for the definition of a positive probability density. We will now turn to the study of some particular solutions of the Dirac equation. This will show that the problem of negative energies is still present in the spectrum of the Dirac equation. However, we will also find that the Dirac equation allows us to fundamentally reinterpret our understanding of *spin*.

3.4.1 Plane wave solutions of the Dirac equation

We will start by looking for some of the simplest solutions to the Dirac equation: plane wave solutions. We make the following Ansatz,

$$\Psi(\underline{r},t) = Au(p)e^{\frac{i}{\hbar}\left(\underline{p}\cdot\underline{r}-Et\right)} = Au(p)e^{-\frac{i}{\hbar}p_\mu x^\mu} \ . \tag{3.47}$$

A is an unknown complex constant, $u(p)$ is a 4-component spinor which is independent of the space-time coordinate x_μ (but depends on the 4-momentum, p_μ).

We insert this Ansatz into the Dirac equation to find the necessary relationships between p_μ and $u(p)$ for this to be a solution. We find

$$\begin{aligned} & i\hbar\frac{\partial\Psi}{\partial t} = \left(-i\hbar c\alpha^k\nabla_k + \beta mc^2\right)\Psi(\underline{x},t), \\ \Longrightarrow \qquad & Eu(p) = \left(c\alpha^k p_k + \beta mc^2\right)u(p), \end{aligned}$$

where we have cancelled the constant A and the overall exponential factor in the line above.

As we are working in a Lorentz covariant system, we can choose our frame to make the problem simpler. Let's start by choosing a frame in which the particle is at rest,

$$p^\mu = (m,0,0,0).$$

In this case, we find,

$$Eu(0) = mc^2\beta u(0) = \begin{pmatrix} mc^2 I_2 & 0 \\ 0 & -mc^2 I_2 \end{pmatrix} u(0), \tag{3.48}$$

in 2×2 block notation, or in 4×4 notation

$$E \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} = \begin{pmatrix} mc^2 & 0 & 0 & 0 \\ 0 & mc^2 & 0 & 0 \\ 0 & 0 & -mc^2 & 0 \\ 0 & 0 & 0 & -mc^2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}.$$

We can see that this solution has two types of solution. If $E = mc^2$ then the components u_1 and u_2 are allowed to be non-zero. If $E = -mc^2$ then only the components u_3 and u_4 can be non-zero. This second class of solution is a little troubling: we again have negative energy solutions. We will return to the interpretation of these solutions later, but for now we continue discussing the plane wave solutions in general.

The Dirac plane wave solutions in the particle's rest-frame have two different energies $E = \pm mc^2$, and for each energy there is a 2-dimensional set of solutions. We can therefore describe any solution to the Dirac equation with a basis of 4 spinors,

$$u^{(1)}(0) = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u^{(2)}(0) = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad u^{(3)}(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad u^{(4)}(0) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{3.49}$$

Although I have not proved it: the first two spinors are associated with positive energies and describe the two spin states of a spin 1/2 particle (*e.g.* the electron). The second two spinors are associated with negative energies and describe the two spin states of a spin 1/2 *anti-particle* (*e.g.* the positron).

We will discuss antimatter further when we return to consider the interpretatons of the Dirac equation in Lecture 17. For now, it suffices to note that there are four degrees of freedom in the solutions to the Dirac equation. The most general solution in the particle's restframe can therefore be expanded in the basis spinors given above,

$$u(0) = c_1 u^{(1)}(0) + c_2 u^{(2)}(0) + c_3 u^{(3)}(0) + c_4 u^{(4)}(0).$$

To find the solutions for a finite momentum, we can perform a Lorentz boost on this spinor. For the system with finite momentum $\underline{p}$ in the x-direction

$$\Psi(\underline{x}, t) = A u(p) e^{-\frac{i}{\hbar} p_\mu x^\mu}, \tag{3.50}$$

where the new constant spinor is given by

$$
\begin{aligned}
u(p) &= S(\Lambda)u(0) \\
&= c_1 \begin{pmatrix} \cosh\frac{\eta}{2} \\ 0 \\ 0 \\ \sinh\frac{\eta}{2} \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ \cosh\frac{\eta}{2} \\ \sinh\frac{\eta}{2} \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ \sinh\frac{\eta}{2} \\ \cosh\frac{\eta}{2} \\ 0 \end{pmatrix} + c_4 \begin{pmatrix} \sinh\frac{\eta}{2} \\ 0 \\ 0 \\ \cosh\frac{\eta}{2} \end{pmatrix} .
\end{aligned}
$$

3.4.2 General solutions of the Klein-Gordon and the Dirac equations

The general solution of the Klein-Gordon equation is given by,

$$
\Phi(\underline{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} \left(a_p e^{-ip_\mu x^\mu} + b_p^* e^{ip_\mu x^\mu} \right) |_{p^0=\omega_p}, \tag{3.51}
$$

where $\omega_p = +\sqrt{\underline{p}^2 + m^2}$.

$\Phi(\underline{x}, t)$ is a complex-valued classical field. The free Klein-Gordon equation is a linear second-order differential equation, and hence, unsurprisingly, its general solution written above contains two undetermined coefficients: a_k and b_k^*.

As explained in the lectures, the correct interpretation of $\Phi(\underline{x}, t)$ in quantum field theory is that it is an operator. Correspondingly, the quantities a_k and b_k^* also have the meaning of the annihilation and creation operators acting on the states from the Hilbert space,

$$
\hat{\Phi}(\underline{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} \left(\hat{a}_p e^{-ip_\mu x^\mu} + \hat{b}_p^* e^{ip_\mu x^\mu} \right) |_{p^0=\omega_p}. \tag{3.52}
$$

The operator $\hat{a}_k$ annihilates a particle of with momentum k^μ, while the operator $\hat{b}_k^*$ creates a particle of the same momentum. The electric charges of the particles of a-type and b-type are opposite.

Now, since every solution of the Dirac equation must automatically be also a solution of the Klein-Gordon equation, we search for the general solution of the Dirac equation in the form,

$$
\hat{\Psi}(\underline{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} \sum_{i=1,2} \left(u_i(p) \hat{a}_{p,i} e^{-ip_\mu x^\mu} + v_i(p) \hat{b}_{p,i}^* e^{ip_\mu x^\mu} \right) |_{p^0=\omega_p}, \tag{3.53}
$$

where $u_i(p)$ and $v_i(p)$ are spinors. By requiring that the Dirac equation is satisfied it immediately follows that these spinors solve the Dirac equations in the momentum space:

$$(\gamma^\mu p_\mu - m)u_i(p) \,=\, 0\,, \qquad (\gamma^\mu p_\mu + m)v_i(p) \,=\, 0\,, \tag{3.54}$$

and there are two solutions for each of these equations, labeled by $i = 1, 2$.

$\hat{\Psi}$ is an operator-valued quantum field of the Dirac fermion, and the operators $\hat{a}^*_{p,i}$ and $\hat{b}^*_{p,i}$ create fermions with the opposite electric charges (electrons and positrons) from the vacuum; the 4-momentum of these particles is p^μ and $i = 1, 2$ labels two different spin projections fo the spin 1/2 particle.

The Hermitian-conjugate operators $\hat{a}_{p,i}$ and $\hat{b}_{p,i}$ annihilate the same particles.

3.4.3 Non-relativistic limit of the Dirac equation

The Dirac equation has solutions which multiple components, and I have told you without proof that these are related to the different spin states of the fermion. We can investigate this further by looking at the non-relativisitc limit of the Dirac equation. In this case, we would expect to recover the Schrödinger equation, just as we did for the Klein-Gordon equation. However, as we will see, the multiple components leads to the incoporation of spin in a natural way.

As with the Klein-Gordon equation, to make things more interesting we discuss the non-relativistic limit of the Dirac equation with coupling to an electromagnetic field. We do this via the minimal coupling procedure,

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[c\alpha^k\left(p_k - eA_k\right) + \beta mc^2 + e\Phi\right]\Psi, \tag{3.55}$$

where A_k denotes the k-th component of the electromagnetic 3-vector potential $\underline{A}$ and Φ is the electromagnetic scalar potential.

Next we decompose the four-component Dirac-spinor Ψ into two two-component spinors $\tilde{\phi}$ and $\tilde{\chi}$. This will simplify the analysis, and is equivalent to working in 2×2 block notation for the Dirac matrices,

$$\Psi = \begin{pmatrix} \tilde{\phi} \\ \tilde{\chi} \end{pmatrix}, \tag{3.56}$$

where $\tilde{\phi} = (\tilde{\phi}_1, \tilde{\phi}_2)^{\mathrm{T}}$ and $\tilde{\chi} = (\tilde{\chi}_1, \tilde{\chi}_2)^{\mathrm{T}}$. Inserting this into Eq. (3.55) we find

$$i\hbar\frac{\partial}{\partial t}\begin{pmatrix}\tilde{\phi}\\ \tilde{\chi}\end{pmatrix} = \begin{pmatrix}\left[e\Phi + mc^2\right] I_2 & c\underline{\sigma}\cdot\underline{\pi}\\ c\underline{\sigma}\cdot\underline{\pi} & \left[e\Phi - mc^2\right] I_2\end{pmatrix}\begin{pmatrix}\tilde{\phi}\\ \tilde{\chi}\end{pmatrix}.$$

where we have used the explicit form of the α^k and β matrices and introduced the operator for the canonical momentum to simplify things

$$\underline{\pi} = \underline{p} - e\underline{A}\,. \tag{3.57}$$

In the same way that we did for the Klein-Gordon equation, we note that the energy of a relativistic particle does not correspond exactly to the non-relativisitc energy in the non-relativisitc limit. The relativisitic energy will have a large contribution from the rest mass of the particle,

$$E = mc^2 + \frac{1}{2}mv^2 + \mathcal{O}\left(\beta^4\right).$$

Therefore, we redefine our wavefunctions to remove this rest mass energy dependence,

$$\begin{pmatrix}\tilde{\phi}\\ \tilde{\chi}\end{pmatrix} =: e^{-\frac{i}{\hbar}mc^2}\begin{pmatrix}\phi\\ \chi\end{pmatrix}.$$

Plugging this into our matrix expression for the Dirac equation, we find

$$\begin{aligned} i\hbar\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \chi\end{pmatrix} + mc^2\begin{pmatrix}\phi\\ \chi\end{pmatrix} &= \begin{pmatrix}\left[e\Phi + mc^2\right] I_2 & c\underline{\sigma}\cdot\underline{\pi}\\ c\underline{\sigma}\cdot\underline{\pi} & \left[e\Phi - mc^2\right] I_2\end{pmatrix}\begin{pmatrix}\phi\\ \chi\end{pmatrix},\\ \implies \qquad i\hbar\frac{\partial}{\partial t}\begin{pmatrix}\phi\\ \chi\end{pmatrix} &= \begin{pmatrix}e\Phi I_2 & c\underline{\sigma}\cdot\underline{\pi}\\ c\underline{\sigma}\cdot\underline{\pi} & \left[e\Phi - 2mc^2\right] I_2\end{pmatrix}\begin{pmatrix}\phi\\ \chi\end{pmatrix}. \end{aligned} \tag{3.58}$$

If we assume that in the non-relativistic limit the following constraints hold,

$$i\hbar\frac{\partial\chi}{\partial t} \ll 2mc^2\chi \qquad \text{and} \qquad e\Phi\chi \ll 2mc^2\chi.$$

Then the second row of Eq. (3.58) allows us to find an expression for χ in terms of ϕ,

$$\begin{aligned} \chi &= \frac{\underline{\sigma}\cdot\underline{\pi}}{2mc}\phi,\\ &\approx \mathcal{O}\left(\frac{v}{c}\right)\phi. \end{aligned} \tag{3.59}$$

In the last line we estimate the size of the quantity in front of ϕ. This is essentially the momentum over the energy, which is proportional to the partice's velocity. Therefore, the component χ is expected to be much smaller than the component ϕ in the non-relativistic limit $v \ll c$.

If we use Eq. (3.59) to eliminate χ in Eq. (3.58), we find the top row gives us and equation for ϕ

$$\begin{aligned} i\hbar\frac{\partial\phi}{\partial t} &= c\underline{\sigma}\cdot\underline{\pi}\chi + e\Phi\phi, \\ &= \frac{(\underline{\sigma}\cdot\underline{\pi})^2}{2m}\phi + e\Phi\phi. \end{aligned}$$

We can evaluate the square of the operator $\underline{\sigma}\cdot\underline{\pi}$ explictly,

$$\begin{aligned} (\underline{\sigma}\cdot\underline{\pi})^2 &= \sigma_i\sigma_j\pi_i\pi_j, \\ &= \frac{1}{2}\left\{\sigma_i,\sigma_j\right\}\pi_i\pi_j + \frac{1}{2}\left[\sigma_i,\sigma_j\right]\pi_i\pi_j, \\ &= \delta_{ij}\pi_i\pi_j + i\varepsilon_{kij}\sigma_k\pi_i\pi_j, \\ &= \underline{\pi}^2 + i\underline{\sigma}\cdot\left(\underline{\pi}\times\underline{\pi}\right), \end{aligned}$$

which we can further simplify by expanding the product $\underline{\pi}\times\underline{\pi}$,

$$\begin{aligned} (\underline{\pi}\times\underline{\pi})_i &= \varepsilon_{ijk}\left(p_j - eA_j\right)\left(p_k - eA_k\right), \\ &= -e\varepsilon_{ijk}\left(p_jA_k + A_jp_k\right), \\ &= -e\varepsilon_{ijk}\left[-i\hbar\frac{\partial A_k}{\partial x^j}\right], \\ &= i\hbar e\left(\underline{\nabla}\times\underline{A}\right)_i = i\hbar eB_i. \end{aligned}$$

This is just the magnetic field! Putting this all together, we have found

$$(\underline{\sigma}\cdot\underline{\pi})^2 = \underline{\pi}^2 - e\hbar\underline{\sigma}\cdot\underline{B},$$

and therefore,

$$i\hbar\frac{\partial\phi}{\partial t} = \left[\frac{\underline{\pi}^2}{2m} - \frac{e\hbar}{2m}\underline{\sigma}\cdot\underline{B} + e\Phi\right]\phi.$$

This is known in non-relativistic quantum mechanics as the *Pauli equation*, and it was original derived in 1927 (a year before the Dirac equation was first proposed) as a modification of the Schrödinger equation coupled to an electromagnetic field which accounts for the spin of a spin-$\frac{1}{2}$ particle.

This is a remarkable accomplishment for the Dirac equation. We started out by trying to write down a first-order wave equation consistent with the relativistic energy-momentum relation

$$E^2 = \underline{p}^2 c^2 + m^2 c^4,$$

and this has led us to an equation which naturally incorporates spin!

In fact, it goes further. If we look at the case of a constant $\underline{B}$ field,

$$\underline{A} = \frac{1}{2}\underline{B} \times \underline{x} \qquad \text{and} \qquad \underline{B} = \underline{\nabla} \times \underline{A},$$

and introduce spin and orbital angular momentum operators,

$$\underline{S} = \frac{\hbar}{2}\underline{\sigma} \qquad \text{and} \qquad \underline{L} = \underline{x} \times \underline{p},$$

we can re-express the Pauli equation as

$$i\hbar\frac{\partial\phi}{\partial t} = \left[\frac{\underline{p}^2}{2m} + \frac{e^2}{2m}\underline{A}^2 - \frac{e}{2m}\left(\underline{L} + 2\underline{S}\right) \cdot \underline{B} + e\Phi\right]\phi.$$

In this equation the factor of 2 in front of $\underline{S}$ is known as the Landé g-factor g. We see that the Dirac equation has predicted that

$$g = 2.$$

Up until that point, this quantity had to be measured in experiment, and it was known that it took a value close to 2. Not only has the Dirac equation accounted for spin, but it has also led to an explanation of the size of the Landé g-factor. Again, at the risk of repeating myself, I'll stress that this new understanding of spin has emerged from our attempts to reconcile special relativity with quantum mechanics. This is an unexpected and truly remarkable discovery and it's well worth boring your students with whenever you get the chance.

The Dirac equation predicts $g = 2$ exactly. This is in good agreement with experimental data, but we now know that the real value is slightly higher than 2. Experimentally, we know that for the electron the *anomalous magnetic moment* is given by

$$a_e = \frac{g_e - 2}{2} = 0.00115965218073(28).$$

where the subscript just denotes that this is for the electron. This small deviation from the predictions of the Dirac equation can be explained by

Quantum Field Theory (QFT). Indeed, the state of the art computation of this number using all the advanced mathematical machinery of QFT gives us the theoretical prediction of

$$a_e = 0.00115965218178(77).$$

This is an exceptionally good agreement; it is one of the most precisely tested quantities in all of science.

3.5 Solutions of the Dirac equation for a central potential

Core reading: Chapter 15.5 of Bransden & Joachain

Central potentials are common features in physical models, we saw in the treatment of scattering that they are often a good model of particle interactions. One of the most important appearances of the central potential is of course in spectroscopy and the computation of energy levels for an atomic system. This is something you are familiar with from non-relativisitic quantum mechanics. Indeed, you already know that the energy levels of Hydrogren are given in the Schrödinger equation to be

$$E_n = -\frac{\mathrm{Ry}}{n^2}.$$

where Ry is the *Rydberg constant* Ry = 13.6 eV. This result is in reasonable agreement with known atomic spectra, but we know that it can be improved by treating other effects via pertubation theory. Incorporating relativistic corrections and spin-orbit coupling terms allows the *fine structure* of the Hydrogen spectrum to be computed. See for example, chapters 7.5 and 8.2 in Bransden & Joachain.

In this section we will see how to solve the Dirac equation in a central potential. This will allow us to compute the spectrum of Hydrogen with all relativistic effects already incorporated.

3.5.1 Angular momentum in a central potential

We start from the Dirac equation minimally coupled to the electromagnetic field.

$$i\hbar\frac{\partial\Psi(\underline{r},t)}{\partial t} = \left[c\alpha^k p_k + \beta mc^2 + e\Phi\right]\Psi(\underline{r},t)\ , \tag{3.60}$$

we assume that there is no magnetic field $\underline{A} = 0$ and that the electromagnetic scalar potential is central,

$$e\Phi := V(r).$$

We focus on stationary solutions, $\Psi(\underline{r},t) = e^{-i\frac{E}{\hbar}t}\Psi(\underline{r})$, for which the Dirac equation reads,

$$E\Psi(\underline{r}) = \left[c\alpha^k p_k + \beta mc^2 + V(r)\right]\Psi(\underline{r})\ . \tag{3.61}$$

We would like to look for separable solutions, just as we would have done for the Schrödinger equation. In the non-relativisitc case, the operators $\underline{L}^2$

and L_z commuted with the Hamiltonian, and therefore corresponded to conserved quantum numbers. This allowed us to focus on solutions that were simultaneous eigenvectors of H, $\underline{L}^2$ and L_z. Things are not quite so simple for the Dirac equation. We can see this by computing the commutator of the Dirac Hamiltonian in a central potential

$$H_{\mathrm{D}} = c\alpha^k p_k + \beta mc^2 + V(r),$$

and the orbital angular momentum operator $\underline{L} = \underline{r} \times \underline{p}$,

$$\begin{aligned}
[H_{\mathrm{D}}, L_m] &= c\alpha^k \left[p_k, (\underline{r} \times \underline{p})_m\right] + \left[\beta mc^2, (\underline{r} \times \underline{p})_m\right] + \left[V(r), (\underline{r} \times \underline{p})_m\right], \\
&= c\varepsilon_{ijm}\alpha^k \left[p_k, r_i p_j\right] + \varepsilon_{ijm}[V(r), r_i p_j], \\
&= c\varepsilon_{ijm}\alpha^k \left(r_i \left[p_k, p_j\right] + \left[p_k, r_i\right] p_j\right) + \varepsilon_{ijm}[V(r), r_i p_j], \\
&= -ic\hbar\varepsilon_{ijm}\alpha^k \delta_{ki} p_j + \varepsilon_{ijm}[V(r), r_i p_j], \\
&= -ic\hbar \left(\underline{\alpha} \times \underline{p}\right)_m + \varepsilon_{ijm} r_i [V(r), p_j], \\
&= -ic\hbar \left(\underline{\alpha} \times \underline{p}\right)_m + i\hbar\varepsilon_{ijm} r_i \frac{\partial V}{\partial r^j}, \\
&= -ic\hbar \left(\underline{\alpha} \times \underline{p}\right)_m + i\hbar \left(\underline{r} \times \underline{\nabla} V\right)_m .
\end{aligned}$$

The second term in this expression is actually zero: working in cartesian coordinates, it is easy to show that the vector derivative of a central potential is in the radial direction,

$$\underline{\nabla} V = \frac{1}{r}\frac{\partial V}{\partial r}\underline{r},$$

and therefore the cross product vanishes $\underline{r} \times \underline{r} = 0$. Ultimately, we find

$$[H_{\mathrm{D}}, \underline{L}] = -i\hbar c(\underline{\alpha} \times \underline{p}) \neq 0.$$

This might not come as a surprise to you. If you remember the treatment of spin in the case of the hydrogen atom, on the inclusion of this intrinsic quantity of angular momentum, we had to describe states in terms of the *total angular momentum* operator $\underline{J} = \underline{L} + \underline{S}$. As the Dirac equation incorporates spin, the same is true for the Dirac spectrum of a central potential.

We start by defining the spin operator,

$$\underline{S} = \frac{\hbar}{2}\underline{\Sigma} ,$$

with

$$\Sigma_k = \begin{pmatrix} \sigma_k & 0 \\ 0 & \sigma_k \end{pmatrix} .$$

In Problem 7, you will verify the following properties of this operator. Firstly, it obeys the spin commutation relations,

$$[S_i, S_j] = i\hbar\epsilon_{ijk}S_k \ .$$

Secondly, the total spin operator $\underline{S}^2$ is proportional to the identity,

$$\underline{S}^2\Psi \quad = \quad \frac{3}{4}\hbar^2\Psi = s(s+1)\hbar^2\Psi \ ,$$

with $s = 1/2$, as expected for the spin-$\frac{1}{2}$ Dirac equation.

Finally, you will also show that the spin operator does not commute with the Hamiltonian. See the solutions to the Workshop 8 problems for more discussion of how to prove this.

$$[H, \underline{S}] = +i\hbar c(\underline{\alpha} \times \underline{p}) \neq 0 \ .$$

Using these facts, we can verify that the total angular momentum operator $\underline{J}$ does indeed commute with the Dirac Hamiltonian,

$$[H, \underline{J}] = 0 \ .$$

and therefore, we also have

$$[H, \underline{J}^2] = 0 \ .$$

We conclude that we can describe the energy eigenstates of Dirac Hamiltonian in a central potential in terms of states with definite values of $\underline{J}^2$ and J_z, the total angular momentum in the z-direction,

$$\underline{J}^2\Psi = \hbar^2 j(j+1)\Psi \qquad \text{and} \qquad J_z\Psi = \hbar m_j\Psi.$$

3.5.2 Ansatz for a central potential

Unfortunately, finding the form of the solutions for the Dirac equation in a central potential is rather complicated. In this section, we will consider an Ansatz for the solutions, and see how we would go about solving the equations; however, the treatment will not go into all the necessary mathematical detail.[6]

[6] You can see ta full treatment for Hydrogenic central potentials in the suggested chapters of Bransden & Joachain.

We make the following ansatz for the Dirac-equation with a spherical potential

$$\Psi^l_{jm_j}(r,\theta,\phi) = \begin{pmatrix} \frac{iG_{lj}(r)}{r} & & \phi^l_{jm_j}(\theta,\phi) \\ \frac{iF_{lj}(r)}{r} & \left(\underline{\sigma}\cdot\frac{\underline{x}}{x}\right) & \phi^l_{jm_j}(\theta,\phi) \end{pmatrix}$$

with the 2-component *spherical spinors* defined by

$$\phi^l_{jm_j}(\theta,\phi) = \begin{cases} \phi^{(+)}_{jm_j}(\theta,\phi) & \text{for} \quad j = l+\frac{1}{2} \\ \phi^{(-)}_{jm_j}(\theta,\phi) & \text{for} \quad j = l-\frac{1}{2} \end{cases}$$

and

$$\phi^{(+)}_{jm_j}(\theta,\phi) = \begin{pmatrix} \sqrt{\frac{l+m_j+\frac{1}{2}}{2l+1}}Y_{l,m_j-\frac{1}{2}}(\theta,\phi) \\ \sqrt{\frac{l-m_j+\frac{1}{2}}{2l+1}}Y_{l,m_j+\frac{1}{2}}(\theta,\phi) \end{pmatrix} ,$$

$$\phi^{(-)}_{jm_j}(\theta,\phi) = \begin{pmatrix} \sqrt{\frac{l-m_j+\frac{1}{2}}{2l+1}}Y_{l,m_j-\frac{1}{2}}(\theta,\phi) \\ -\sqrt{\frac{l+m_j+\frac{1}{2}}{2l+1}}Y_{l,m_j+\frac{1}{2}}(\theta,\phi) \end{pmatrix} .$$

Y are the spherical harmonics. You will explore the properties of these spherical spinors in Workshop 8, problem 8.1. They are analogous quantities to the spherical harmonics for a spin-$\frac{1}{2}$ system. We have also introduced $G_{lj}(r)$ and $F_{lj}(r)$ to describe the dependence on the radial distance r. We can find a differential equation that these functions must satisfy by plugging our Ansatz into the Dirac equation,

$$\begin{aligned} \frac{E-m-V(r)}{\hbar c}G_{lj}(r) &= -\frac{dF_{lj}(r)}{dr} \mp \left(j+\frac{1}{2}\right)\frac{F_{lj}(r)}{r} \\ \frac{E+m-V(r)}{\hbar c}F_{lj}(r) &= +\frac{dG_{lj}(r)}{dr} \mp \left(j+\frac{1}{2}\right)\frac{G_{lj}(r)}{r} \end{aligned}$$

for $j = l \pm 1/2$. These radial equations can be solved by standard techniques, such as the series method; however, in most situations there will not be a simple closed form expression for these functions.

For a hydrogen-like central potential $V(r) = -Z\alpha/r$, where $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$ is the fine structure constant, it can be shown that we ultimately find the following energy eigenvalues

$$E_{n,j} = \frac{mc^2}{\sqrt{1+\left(\frac{Z\alpha}{n-\left(j+\frac{1}{2}\right)+\sqrt{\left(j+\frac{1}{2}\right)^2-(Z\alpha)^2}}\right)^2}} .$$

This result shows that the Dirac equation predicts exact degeneracy between levels with the same n and j but with different values of J_z and $\underline{L}^2$. If we expand this quantity in powers of $(Z\alpha)$, we get

$$E_{n,j} = mc^2 \left[1 - \frac{Z^2\alpha^2}{2n^2} - \frac{(Z\alpha)^4}{2n^3} \left(\frac{1}{j+\frac{1}{2}} - \frac{3}{4n} \right) + \mathcal{O}\left(Z^6\alpha^6\right) \right] .$$

The first term is just the rest mass and the second term is the result that you would have expected from your knowledge of the Schrödinger theory for the hydrogen atom. The third term is the first relativistic correction, that gives the *fine structure* of Hydrogen-like atoms. You have previously computed this splitting in non-relativisitc quantum mechanics by using perturbation theory on some physically motivated additional terms to the Hamiltonian. We see that these corrections to the Schrödinger Hamiltonian come directly from the Dirac equation which naturally incorporates both relativitiy and spin.

We have noted that the Dirac equation predicts exact degenercy between states of different $\underline{L}^2$ but identical n and j values. In fact, there is a further splitting of the energy levels that cannot be explained by the Dirac equation: states of different value of the $\underline{L}^2$ operator are split by a small amount. This is known as the *Lamb shift*. To explain this splittng, we must once again turn to Quantum Field Theory, where fluctuations in the quantum fields lead to small additional corrections to the energy spectrum.

3.6 Negative energies and Dirac's hole theory

Core reading: Chapter 17.1 of Bransden & Joachain

(Highly recommended reading: Chapter 10 of Advanced Quantum Mechanics by Schwabl)

In this lecture I presented a more qualitative discussion of the negative energy states associated with the Dirac equation. Although the core concepts are on syllabus, the mathematical detail is extension material, and as such I will not provide a typed version of it. However, here is a brief summary of the material covered:

- We discussed how a fermion coupled to an electromagnetic field with accessible negative energy states would be able to decay by the emission of a photon.
- We discussed how the negative energy components appear when a particle is localized and the role they can play in scattering problems.
- We then discussed Dirac's hole theory: that the negative energy levels are simply occupied by fermions already, and that the Pauli exclusion principle then forbids a positive energy particle from decaying into these states. This predicts states with the opposite charge to the original fermion, so-called anti-particles. The anti-electron is known as the positron and denoted e^+.
- Dirac's hole theory also predicts *particle number changing processes*:
  - Particle production, where a photon turns into a fermion and an anti-fermion: $\gamma \to e^+e^-$.
  - Particle annihilation, where matter and anti-matter combine into photons: $e^+e^- \to \gamma$.
- We discussed how these number changing processes hint at the ultimate resolution of the problems seen so far. They arise due to the interpretation of the Dirac equation as a *single particle* wave equation. In the formulation of Quantum Field Theory this condition is relaxed, and processes with variable particle number are allowed.

Chapter 4

Measurement and the Intepretation of Quantum Mechanics

4.1 Quantum measurement and the interpretation of quantum mechanics

Core reading: Chapter 17.1 to 17.4 of Bransden & Joachain

I realise that I run the risk of hyperbole, but in my opinion Quantum Mechanics (QM) is one of the most startling developments in the history of science. On par with general relativity, it has fundamentally changed how the world is seen and our conception of the structure of a fundamental theory of nature. However, despite its paradigm-busting revision of some of the basic questions of science, it is also, in some sense, *true*. Its predictions are proven everyday in modern technological application: the laser, semiconductors, high-precision clocks ... Not to mention the countless times it has been tested in the laboratory. It is a key part of modern theories of the electron and photon, and we have already seen some of the successes of this theory, for example when measuring the magnetic moment of the electron,

$$\frac{g_e - 2}{2} = 0.00115965218\ldots.$$

We can therefore safely conclude that something about QM is 'correct'. That our current formulation of its rules appears to at least well reproduce the microscopic behaviour of nature, at least at energies and distances that we can probe today.

Despite its successes, the history of quantum mechanics is one of contentious debate about the ultimate *interpretation* of the theory. The fundamental role played by inherently probabilistic quantities continues to cause many to doubt that it is a final theory. To some extent, any debate on the interpretation of the theory is a *philosophical* one. Provided you never change the predictions of QM, you can never hope to establish your interpretation scientifically. This has led many to dismiss the issue, and the famous quip *"Shut up and calculate!"* usually shadows any debate in the physics literature. However, the origin of this quotation shows that not all physicists subscribe to such a pragmatic viewpoint:

> *"If I were forced to sum up in one sentence what the Copenhagen interpretation says to me, it would be "Shut up and calculate!"*
>
> – David Mermin, *Physics Today* April 1989

We will end this lecture course by discussing the foundations of quantum mechanics and some of the most contentious points of its interpretation.

We will then study a modified class of theories, so-called *hidden variable* theories, and see how these in fact *do* lead to measurable differences in the form of *Bell's inequalities*.

4.1.1 The basic question

We know that QM has an inherently probabilistic character inherited from the Born interpretation of mod-square amplitudes as probabilities. The inherentness of this randomness was a major source of debate, and led to the formulation of hidden variable theories which replace inherent with statistical randomness.

To understand these issues, let us start by recapitulating the basics of quantum mechanics:

1. Observables are described by hermitian operators.
   - An observable A is related to a hermitian operator $\hat{A}$.
   - Hermitian operators have a complete set of eigenstates $|n\rangle$ with
   $$\hat{A}|n\rangle = a_n|n\rangle \ .$$
   - The real eigenvalues of this operator a_n are the possible outcomes of a measurement of the observable A.
2. Quantum states are described by wave functions. A general wave function $|\Psi\rangle$ can be written as a linear combination of the $|n\rangle$:
   $$|\Psi\rangle = \sum_n c_n|n\rangle \ .$$
3. The time evolution of the system is governed by the Hamiltonian operator
   $$i\hbar\frac{\partial}{\partial t}\,|\Psi\rangle = \hat{H}\,|\Psi\rangle \, .$$
4. *(Born's interpretation)* The probability of measuring the state $|\Psi\rangle$ to have value a_n of the observable A is given by the coefficient mod-squared in front of $|n\rangle$,
   $$P\left(A = a_n\right) = |c_n|^2 \, .$$
5. If we measure the observable A in a system described by $|\Psi\rangle$ and we get the result a_n then the wave function *collapses* to the state $|n\rangle$. (A corollary of this point is that repeated measurements would be expected to observed the same value of A.)

The two most contentious axioms are probably 4 and 5. Axiom 4 introduces the inherent probabilistic nature of quantum mechanics. Axiom 5 introduces the notion of collapse: measuring a state fundamentally alters the state itself, losing information about its previous configuration.

To see how this mathematical formulation might leave room for interpretational issues, let us consider a free particle and we want to make a measurement of its position:

- Quantum mechanics gives us only the probability for finding the particle at a certain location at a given time. The probability of finding the particle at time t_0 at the location $\underline{x}_0$ is given by $|\Psi(\underline{x} = \underline{x}_0, t = t_0)|^2$.
- The measurement of the position at time t_0 gave the result $\underline{x}_0$. This means that the original wave function has now collapsed into a position eigenstate with the eigenvalue $\underline{x}_0$ (Axiom 5).
- What can we say at this point about the position of the particle at $t_0 - \delta t$ for some small positive δt?

The answer to this question is not governed by our general rules of quantum mechanics, some possible answers are:

1. at point $\underline{x}_0$ (realistic position);
2. nowhere (orthodox position);
3. there is no answer to this question (agnostic position),

Let's think a bit about these responses:

1. If the particle were actually at the position $\underline{x}_0$ (or some other real position $\underline{x} - \underline{v}\delta t$ etc.) and we just could not describe it with quantum mechanics, this would be a failure of the current formulation: there are relevant physical quantities that it simply does not know. In this case we could expect there to be a more fundamental theory, one that contains all the results from quantum mechanics and in addition knows where the particle was before the measurement.

   Theories such as this are known as hidden variable theories. The extra parameters which are missing from QM are the 'hidden' variables. In this case, the inherent randomness of QM is replaced by a statistical randomness. The analogy is with statistical mechanics. If we consider a gas, we cannot predict the momentum of any single atom

we choose to observe; we could only predict average properties. This is ultimately due to our ignorance, not anything fundamentally random about momentum. Many scientists have found such theories of quantum mechanics an attractive alternative to quantum randomness including A. Einstein and D. Bohm.

2. This is more or less the basis of the famous and widespread *Copenhagen interpretation*, formulated by Born, Bohr and others. The particle simply did not have the property of position before the measurement. It was the measurement itself which forced the particle to choose a certain position, through the collapse of the wave function

3. This is a practical attitude. We can not answer this hypothetical question, because our formulation of quantum mechanics simply does not give us a means to answer it.

4.1.2 Einstein-Podolsky-Rosen paradox

In 1935 Einstein, Podolsky and Rosen (EPR) described a thought experiment designed to argue that a realistic position was superior to the orthodox Copenhagen interpretation[1].

To see their argument, consider the decay of a neutral pion (spin = 0) at time t_0 into an electron and a positron (each spin 1/2)

$$\pi^0 \to e^+ + e^- \ .$$

If the pion is at rest then the electron-positron pair is described by the following wave function (spin = 0)

$$\frac{1}{\sqrt{2}} \left(| \uparrow\downarrow\rangle - | \downarrow\uparrow\rangle \right) \ . \tag{4.1}$$

Next we assume that in some distance x_e of the former position of the pion, the spin of the electron will be measured at the time $t_e > t_0$.
Let us assume that the outcome is spin-up. Thus it is clear that any future measurement of the spin of the positron will give spin-down.

How can this be interpreted? A realist could explain the correlations as the states were produced at the same time from the pion. At this moment, we imagine that each e^+e^- chooses the direction for the spin of the electron (and the positron takes the opposite). In this way, nothing has to happen non-locally, but each fermion carries with it some hidden quantity which

[1] Einstein, Podolsky, Rosen *Phys.Rev.* **47** 777 (1935)

dictates its spin orientation. Performed many times, we cannot necessarily predict the spin orientation of any given electron (we don't know what it agreed at the start), but the common source ensures that it is always opposite that of the positron. The orthodox interpretation would instead proceed through the following steps:

$t_0 < t < t_e$	The electron-positron state is described by Eq.(4.1).
$t = t_e$	The wave function collapses.
$t > t_e$	The electron and positron are in pure spin states.

This point of view suggests that interfering with the electron can alter the state of the positron over *arbitrarily large* distances. This is a *non-local* effect, one which Einstein *et al.* viewed as an impossibility. They referred to it as "spukhafte Fernwirkung" or "spooky action-at-a-distance".

I think it is fair to see why the realistic position was so appealing. However, as we will see in the next section, 30 years later a way to distinguish between these two theories was developed.

4.1.3 Bell's inequality

In 1964 J.S. Bell (1928-1990) was generalising the above experiment and managed to show that there is a measurable difference between the predictions of QM and those of hidden-variable models.

We take the same experimental setup as in the EPR case, with the generalisation that now the spin-detectors can be moved in arbitrary directions. In the EPR case both were measuring the z-component of the spin, while the electron-spin component is measured relative to the vector $\underline{a}$ and the positron-spin component is measured relative to the vector $\underline{b}$. In the end the relevant observable will be the quantum mechanical average of the product of the two spins $P(\underline{a}, \underline{b})$. The case $\underline{a} = \underline{b}$ gives the result of the EPR-case. In the EPR case quantum mechanics told us that the product will always give $-1 \cdot 1$. Thus we have

$$P(\underline{a}, \underline{a}) = -1 \ .$$

If the vectors are anti-parallel $\underline{a} = -\underline{b}$, we will obviously get

$$P(\underline{a}, -\underline{a}) = +1 \ .$$

For a general direction of the vectors quantum mechanics will give

$$P(\underline{a}, \underline{b}) = -\underline{a} \cdot \underline{b} \ .$$

It will turn out, however, that this simple result is incompatible with theories with hidden variables. To show that, we consider very general models with hidden variables, assuming only that reality is given by the quantum mechanical wave function Ψ and some hidden variable λ.

- $A(\underline{a}, \lambda)$ gives the result of the measurement of the electron spin relative to the vector $\underline{a}$. It can take the values ± 1.
- $B(\underline{b}, \lambda)$ gives the result of the measurement of the positron spin relative to the vector $\underline{b}$. It can take the values ± 1.

If $\underline{a} = \underline{b}$, we have

$$A(\underline{a}, \lambda) = -B(\underline{a}, \lambda) \ .$$

The average of the product of the two measurements is given by

$$P(\underline{a}, \underline{b}) = \int \rho(\lambda) A(\underline{a}, \lambda) B(\underline{b}, \lambda) d\lambda \ ,$$

with $\rho(\lambda)$ being the unknown probability distribution of the hidden variable which must satisfy,

$$\begin{aligned} 0 \leq \rho(\lambda) &\leq 1 \ , \\ \int \rho(\lambda) d\lambda &= 1 \ . \end{aligned}$$

Because of $B(\underline{b}, \lambda) = -A(\underline{b}, \lambda)$ we get

$$P(\underline{a}, \underline{b}) = -\int \rho(\lambda) A(\underline{a}, \lambda) A(\underline{b}, \lambda) d\lambda \ .$$

Next we take an additional general unit vector $\underline{c}$ to get

$$\begin{aligned} P(\underline{a}, \underline{b}) - P(\underline{a}, \underline{c}) &= -\int \rho(\lambda) \left[A(\underline{a}, \lambda) A(\underline{b}, \lambda) - A(\underline{a}, \lambda) A(\underline{c}, \lambda) \right] d\lambda \\ &= -\int \rho(\lambda) \left[A(\underline{a}, \lambda) A(\underline{b}, \lambda) - |A(\underline{b}, \lambda)|^2 A(\underline{a}, \lambda) A(\underline{c}, \lambda) \right] d\lambda \\ &= -\int \rho(\lambda) \left[1 - A(\underline{b}, \lambda) A(\underline{c}, \lambda) \right] A(\underline{a}, \lambda) A(\underline{b}, \lambda) d\lambda \ . \end{aligned}$$

Here we used $|A(\underline{b}, \lambda)|^2 = 1$. Next we keep in mind

$$\begin{aligned} \rho(\lambda) \left[1 - A(\underline{b}, \lambda) A(\underline{c}, \lambda) \right] &\geq 0 \ , \\ -1 \leq A(\underline{a}, \lambda) A(\underline{b}, \lambda) &\leq +1 \ , \end{aligned}$$

to get

$$\begin{aligned} |P(\underline{a},\underline{b}) - P(\underline{a},\underline{c})| &\leq \int \rho(\lambda)\,[1 - A(\underline{b},\lambda)A(\underline{c},\lambda)]\,d\lambda \\ &= 1 + P(\underline{b},\underline{c})\ . \end{aligned}$$

This is *Bell's inequality.*[2] It must be held in all theories with hidden variables (provided they obey our assumptions about them).

The remarkable thing is that QM does not satisfy Bell's inequality. We already argued that in QM we have $P(\underline{a},\underline{b}) = -\underline{a}\cdot\underline{b}$. Therefore, Bell's inequality tells us

$$|\underline{a}\cdot\underline{b} - \underline{a}\cdot\underline{c}| \quad\leq\quad 1 - \underline{b}\cdot\underline{c}\ .$$

This relation can easily be violated, assume *e.g.* $\underline{a}$, $\underline{b}$ and $\underline{c}$ lie in a plane, $\underline{a}$ and $\underline{b}$ are perpendicular and $\underline{c}$ lies in between them (with an angle of 45°). Then we get

$$|0 - 1/\sqrt{2}| = 0.707 \quad\leq\quad 1 - 1/\sqrt{2} = 0.29\ ,$$

which is obviously not correct.

We conclude that (local) hidden variable theories do not have identical predictions to QM. There are more complicated variants of hidden variable theories which are still compatible with data, but these generally give up the locality that provides a nice interpretation of the EPR correlations. To date, all experimental tests have shown that Bell's inequality is violated and that QM is correct. It seems that locality does not hold and that there is an instantaneous collapse of the wave function.

Needless to say, we are still not in a position to say we completely understand the interpretation of QM. Hidden variable theories are not entirely abandoned, theorists enjoy cooking up ways to avoid experimental results, but they are looking more unlikely. Quantum measurement and the process of collapse still remains to some extent a puzzle, but I hope this study of Bell's inequalities has convinced you that there may be other ways to probe the fundamentals of quantum mechanics. Ultimately, we will never find measurable features of novel theories unless physicists reflect on what is justified and unjustified in the august interpretations passed down to them and search for ways to test them.

"Autoritätsdusel ist der größte Feind der Wahrheit."
(*"Unthinking respect for authority is the greatest enemy of truth."*)
– A. Einstein, in a letter to J. Winteler (1901)

[2] There are actually a few different relations all known as Bell's inequalities.